

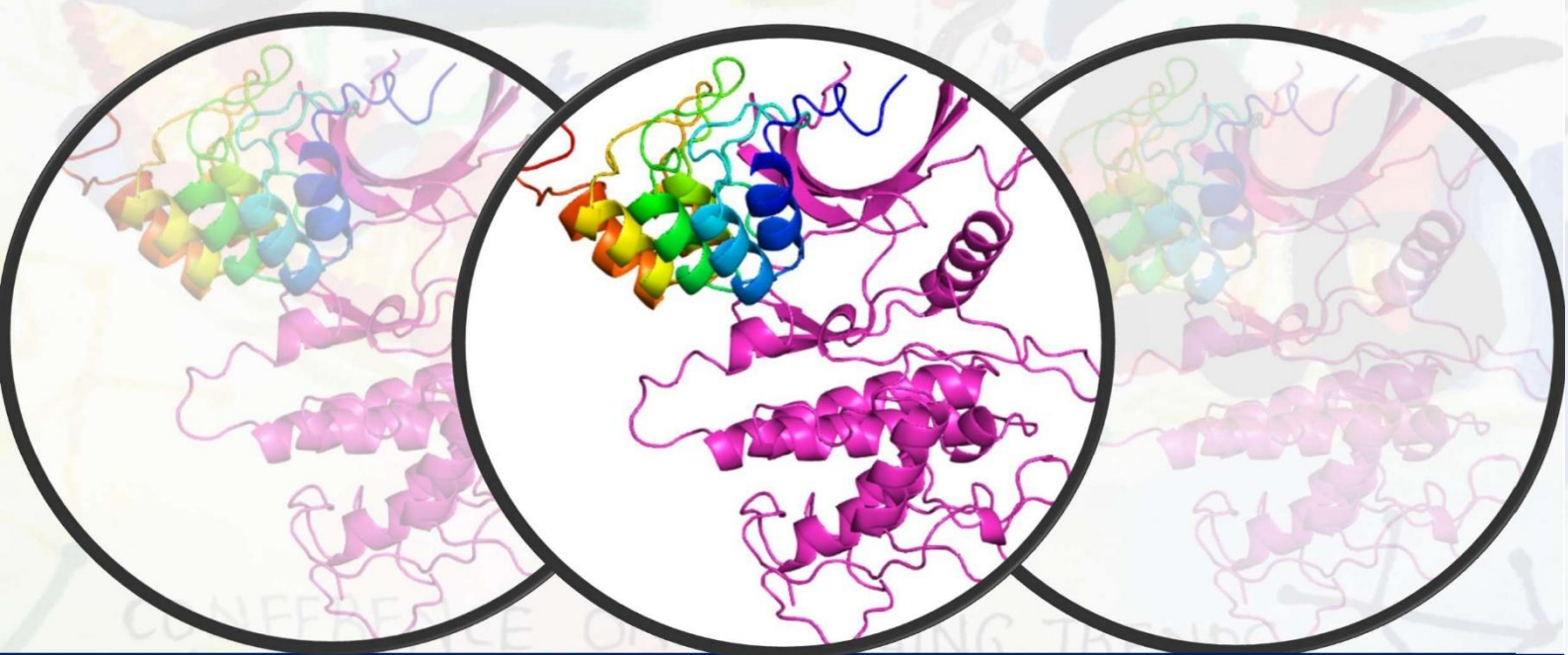
Anusandhan National Research Foundation

And

Department of Biotechnology (DBT), New Delhi sponsored

***National Conference on Emerging Trends in Structural
Biophysics: Integrating Structure, Dynamics and Function
(CETSB 26)***

Abstract Book



JANUARY 9-11, 2026

ORGANIZED BY

DEPARTMENT OF BIOTECHNOLOGY

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL



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CONFERENCE ON EMERGING TRENDS
IN STRUCTURAL BIOPHYSICS



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I am glad that the **Department of Biotechnology** is organizing the **National Conference on Emerging Trends in Structural Biophysics (CETSB-26)** during **January 9–11, 2026**, at the **National Institute of Technology Warangal**. The theme of the conference is very aptly conceived and is both timely and relevant.

Structural biophysics lies at the heart of modern life sciences, seamlessly linking molecular structure with biological function, dynamics, and the fundamental mechanisms governing living systems. Rapid advances in **cryo-electron microscopy, X-ray crystallography, NMR spectroscopy, single-molecule techniques, high-performance computing, and AI-assisted modeling** have revolutionized our ability to probe biomolecular processes at unprecedented resolution. These developments are enabling transformative applications in healthcare, biotechnology, and sustainable solutions. In this context, CETSB-26 provides a timely and valuable platform to deliberate on recent breakthroughs, share cutting-edge research, and identify emerging challenges and future directions in the field.

I am particularly pleased that CETSB-26 coincides with the **celebration of two decades of excellence of the Department of Biotechnology at NIT Warangal**. Over the years, the Department has grown into a vibrant hub of teaching and research, characterized by strong interdisciplinary collaborations and a steadfast commitment to nurturing future-ready graduates and researchers. This conference truly reflects that spirit by bringing together distinguished scientists, early-career researchers, and students from across the country.

As Director of **NIT Warangal**, I firmly believe that the future of scientific advancement lies at the confluence of disciplines, enabled by cutting-edge infrastructure, data-driven approaches, and meaningful collaborations with academia, industry, and national laboratories. The Institute remains committed to fostering such an ecosystem—one that promotes curiosity-driven research while aligning with national priorities in health, biotechnology, sustainability, and technological self-reliance. CETSB-26 exemplifies this institutional vision by encouraging cross-disciplinary dialogue and translating fundamental discoveries into societal impact.

I am confident that the **keynote lectures, invited talks, oral and poster presentations, and technical interactions** will foster meaningful scientific exchange, catalyze collaborative research, and inspire new ideas and directions. I sincerely appreciate the dedicated efforts of the organizing committee, faculty members, staff, student volunteers, and all supporters whose collective contributions have made CETSB-26 possible.

I warmly welcome all delegates to **NIT Warangal** and wish you a **stimulating conference, fruitful discussions, and a pleasant stay on our campus**. I wish the conference a **grand success**.

DIRECTOR



राष्ट्रीय प्रौद्योगिकी संस्थान वरंगल

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Message from the Head of the Department

Department of Biotechnology, NIT Warangal

Celebrating 20 Years of Excellence

It is with great pride and pleasure that I extend a warm welcome to all participants of CETSB-26: National Conference on Emerging Trends in Structural Biophysics, being organised by the Department of Biotechnology, NIT Warangal during **9–11 January 2026**. CETSB-26 is particularly significant as it coincides with the **celebration of 20 years of excellence** of the Department of Biotechnology at NIT Warangal.

Over the past two decades, the Department has grown into a vibrant academic and research ecosystem, contributing meaningfully to quality education, cutting-edge research, and national capacity building in biotechnology and allied disciplines. Structural biophysics has remained a core strength of the Department, underpinning advances in understanding biological mechanisms at the molecular level and enabling translational applications in health, disease, and biotechnology.

CETSB-26 aims to bring together leading scientists, early-career researchers, and students from across the country to deliberate on recent advances in structural biology, biophysics, cryo-electron microscopy, X-ray crystallography, NMR spectroscopy, computational modelling, and integrative approaches. The conference will feature plenary and invited lectures, contributed talks, poster presentations, and focused discussions that reflect both foundational concepts and emerging trends in the field.

The Department of Biotechnology, NIT Warangal remains committed to fostering interdisciplinary research, innovation, and academic leadership. This conference represents our continued effort to create platforms that encourage scientific exchange, nurture young researchers, and strengthen national and international collaborations.

I am confident that CETSB-26 will serve as a memorable scientific event, inspire new research directions, and further consolidate the Department's legacy of excellence. I wish the conference every success and look forward to welcoming all participants to NIT Warangal.

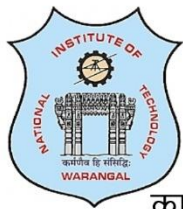
With best wishes,

Prof. Rama Raju B

Head of the Department

Department of Biotechnology

National Institute of Technology Warangal



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Message from the Convenors

It gives us immense pleasure to welcome you to the *National Conference on Emerging Trends in Structural Biophysics (CETSB-26)*, organized by the Department of Biotechnology, NIT Warangal, as we proudly celebrate **20 years of excellence** in teaching, research, outreach, and innovation.

CETSB-26 is envisioned as a platform that brings together scientists, academicians, clinicians, industry partners, and early-career researchers to discuss transformative ideas in structural biology, computational biophysics, cryo-EM, molecular therapeutics, and frontier analytical technologies. Through plenary lectures, keynote talks, poster sessions, and networking events, we aim to foster collaborations that will accelerate scientific discovery and translational impact.

We believe that this conference will not only showcase emerging trends but also inspire future directions for India's scientific leadership in biophysical research. We extend our deepest gratitude to our sponsors, collaborators, advisory committee members, and participants for supporting this initiative.

We wish you an intellectually stimulating and professionally rewarding experience at CETSB-26.

Warm regards,

Prof. Soumya Lipsa Rath

Prof. Thyageshwar Chandran

Convenors, CETSB-26

Department of Biotechnology, NIT Warangal



Programme Schedule

CONFERENCE ON EMERGING TRENDS
IN STRUCTURAL BIOPHYSICS

Time	Session	Speaker	
09.01.2026 Day 1: Modelling and Simulations			
9.00 -10.00	Inaugural		
	High Tea		
10.15-11.00	Prof. G. N. Ramachandran Plenary lecture (Lecture 1)	Prof. Shekhar Mande, Distinguished Professor, Savitribai Phule Pune University's Bioinformatics Centre. Honorary Distinguished Scientist, National Centre for Cell Science (NCCS).	
11.05- 11.50	Prof. M. Vijayan Keynote Lecture (Lecture 2)	Dr. Rajan Sankaranarayanan, CCMB	
11.55-1.25	Lecture 3	Dr. Durba Sengupta, NCL Pune	
1.00 -2.00	Lunch Break		
	Session Chair: Prof. Onkara Perumal		
2.00-2.40	Lecture 4	Prof. Jagannath Mondal, TIFR Hyderabad	Poster Session
2.45-3.25	Lecture 5	Dr. Himanshu Joshi, IIT Hyderabad	
3.30-4.10	Lecture 6	Dr. Aditya Padhi, IIT BHU	
4.15-4.55	Lecture 7 (online)	Dr. Girinath Pillai (Sygnature Discovery, UK)	
4.55 -5.10	Tea Break		
	Session Chair: Prof. Anbumathi		Poster Session
5.10-5.40	Lecture 8	Dr. Suchetana Gupta, Cognizent Life Sciences	
5.40-6.10	Oral presentation Session 1 (4)	Oral presentation Session 2 (4)	
6.10-6.30	Lecture 9	Netweb Technologies	
8.00 pm	Gala Dinner		
10.01.2026 Day 2: Structural Biology			
9.00-9.45	Klaus Schulten Keynote Lecture (Lecture 10)	Prof. Sanjib Senapati, IIT M	
	Session Chair: Prof. Saiket Choudhary		
9.50- 10.20	Lecture 11	Dr. D. Patra, IICB Kolkata	
10.25- 11.10	Lecture 12	Prof. Someswar Rao Sagurthi, JNU	
11.10-11.25	Tea Break		
	Session Chair: Dr. Suchetana Gupta		
11.25-12.10	Lecture 13	Prof. Rajakumara Eerappa, IIT Hyderabad	
12.15-1.00	Lecture 14	Prof. Soumyananda Chakraborti, BITS Hyderabad	

1.00-2.00	Lunch Break		
	Session Chair: Prof. Rajakumara Eerappa		Poster Session
2.00-2.30	Lecture 15	Dr Saikat Choudhury, CCMB	
2.30-3.00	Lecture 16	Prof. Prasenjit Bhaumik, IIT Bombay	
3.05-3.35	Lecture 17	Gator Bio (GeneX)	
3.35-3.45	Tea Break		
	Session Chair: Dr. D. Patra		Poster Session
3.45-4.15	Lecture 18	Prof N. Prakash Prabhu, HCU	
4.15-4.45	Lecture 19	Dr. Prasun, IIT Palakkad	
4.50-5.20	Oral presentation Session 3 (4)	Oral presentation Session 4 (6)	
5.20-5.45	Lecture 20	Dr. Sanchari Banerjee, Sai Life Science, Hyderabad.	
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11.1.2026 Day 3: Bioinformatics			
	Session Chair: Dr. Aditya Padhi		
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9.50- 10.20	Har Govind Khorana Keynote Lecture (Lecture 22)	Dr. Uddhaves, CDAC	
10.25- 11.10	Lecture 23	Dr. Surya Kanta Ghosh, NITW	
11.10-11.25	Tea Break		
	Session Chair: Dr. Prasun		
11.25- 12.10	Lecture 24	Prof. Anshuman Dixit, ILS BBSR	
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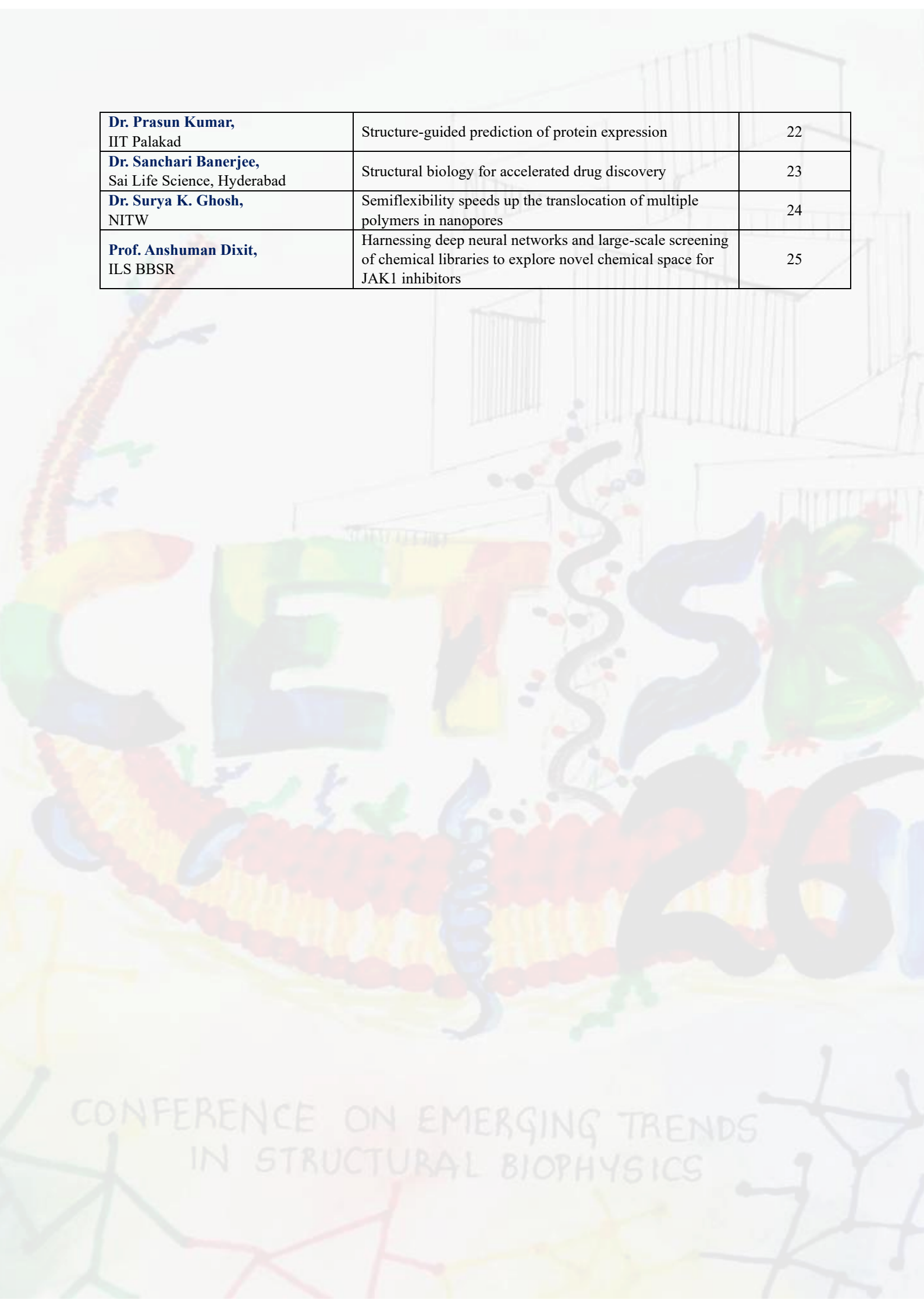


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Keynote speakers

CONFERENCE ON EMERGING TRENDS
IN STRUCTURAL BIOPHYSICS

Structural basis of half-site reactivity in Class Ib Ribonucleotide Reductases

Lumbini Yadav, Shruti Chauhan, Manali Joshi and Shekhar C. Mande*

National Centre for Cell Science, Pune
Bioinformatics Centre, Savitribai Phule Pune University, Pune

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Ribonucleotide reductases (RNRs) catalyse the first committed step in the conversion of ribonucleotides into deoxyribonucleotides, thereby maintaining balanced dNTP pool required for DNA synthesis and repair. Class Ia and Ib RNRs exhibit half-site reactivity, in which only one catalytic subunit within the α_2 dimer is active at a time. We have recently characterized the Class Ib α -subunit from *Mycobacterium thermoresistibile* to investigate how conformational flexibility influences half-site catalysis. Cryo-EM structures of the α -subunit were determined in both the apo form and in a holo complex bound to TTP (effector) and GDP (substrate). These structures reveal significant conformational heterogeneity in several regions, particularly in the N-terminal domain and key regulatory loops surrounding the catalytic and specificity sites. Binding by the effector and substrate appear to stabilize these loops and remodel the catalytic pocket. Isothermal Calorimetry studies show that purine effector nucleotides bind α -subunit with a stoichiometry of one, whereas pyrimidine effector nucleotides bind with a stoichiometry of ~ 0.5 corresponding to one nucleotide per dimer. Despite this asymmetry, both purine and pyrimidine nucleotides bind with micromolar affinities and exhibit predominantly enthalpy-driven thermodynamics. Molecular dynamics simulations, normal mode analysis and 3-dimensional variability analysis reveal coordinated motions in loop 2 in the vicinity of the active site and N-terminal domain, which drive alternating opening and closing of the two active sites. We propose that asymmetric conformational gating mediated by loop 2, loop 4 at the monomer- monomer interface, and the N-terminal region is essential for half-site reactivity. Furthermore, we suggest that conformational sampling by the N-terminal domain is required prior to its engagement with the β -subunit, enabling efficient radical transfer. Together, these findings provide new insights into the structural dynamics and thermodynamic principles that govern regulation and half-site activity in Class Ib RNR regulation.

Chiral proofreaders, their variants and their functional roles

Rajan Sankaranarayanan*

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Despite the presence and biological role of D-amino acids, the fundamental issue of how proteins are made only with the L-chiral entities was largely ignored. Over the last two decades, it has become clearer as to how multiple ‘Chiral Checkpoints’ work in concert to avoid D-amino acids from getting incorporated into proteins (1, 2). Our recent work has shed light on how chiral proofreading systems have played critical roles in important evolutionary transitions (3, 4). In the first part of my talk, I will introduce the work that came out of the laboratories of two of the pioneers of nucleic acid research in the area of protein biosynthesis, Paul Berg and Donald Crothers, more than half a century back. Their work on the identification of D-aminoacyl-tRNA deacylase (DTD) and ‘Discriminator hypothesis’, respectively, were hugely ahead of their time and were partly against the general paradigm at that time. In both of the above works, the smallest and the only achiral amino acid turned out to be an outlier as DTD can act weakly on glycine charged tRNAs with a unique discriminator base of ‘Uracil’. This peculiar nature of glycine remained an enigma for nearly half a century. With a load of available information on the subject by the turn of the century, our work on ‘chiral proofreading’ mechanisms during protein biosynthesis serendipitously led us to revisit these findings. Our analysis has uncovered an unexpected connection between them that has implications for evolution of different eukaryotic life forms (5) and will be the focus of the second part of my talk.

Selected references:

1. Kuncha, S. K. *et al.*, *J. Biol. Chem.* 2019 (Review article).
2. Kumar, P. *et al.* *FEBS Letters* 2022 (Review article).
3. Gogoi, J. *et al.* *Sci. Adv.* 2022.
4. Kumar, P. *et al.* *Proc. Natl. Acad. Sci. (USA)* 2023.
5. Kumar, P. and Sankaranarayanan, R. *NAR* 2024 (Critical Reviews and Perspective article).

An integrative molecular dynamics-machine learning approach to unravel the mechanism of T-Cell immune response and screening of viral epitopes

Sanjib Senapati*

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T-cell immunity is a crucial defense against viral infections in vertebrates. During viral entry, innate immune cells degrade the viral proteins (antigens) and present them *via* Major Histocompatibility Complex (MHC) molecules. T-cell receptors (TCRs) recognize these MHC-bound viral peptides (pMHC), initiating a T-cell-mediated immune response. Despite its significance, the mechanism by which pMHC-TCR binding triggers T-cell activation remains unclear. In this study, we employed an integrative computational approach combining Bioinformatics, Molecular Dynamics (MD) simulations, and Machine Learning (ML) to identify viral epitopes as potential vaccine candidates. We performed large-scale all-atom and coarse-grained MD simulations on MHC-peptide-TCR complexes embedded into dendritic and T-cells, for which experimental immunogenicity data are available. One hundred fifty such systems are simulated for 1 μ s each to capture the conformational and dynamical changes that underlie T-cell activation. Our ML models, trained on simulation-derived structural and dynamical features extracted from 2500 time points, revealed key determinants responsible for T-cell activation with an accuracy of approximately 85%. Notably, we have identified the bending of the TCR alpha membrane region and the buried surface area between pMHC and TCR as critical factors influencing immune response initiation. Our approach unravels the mechanism of T-cell mediated immune response and helps ML-guided screening of viral epitopes for vaccine development.

Structural bioinformatics approaches for understanding and predicting the binding affinities of protein complexes and their mutants

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The interactions of proteins with other molecules are important for several biological functions (1). The strength of these interactions is accounted with binding affinity. Further, mutation of amino acids in the interfaces of protein complexes alter their structure, binding affinity and function, and some of them lead to diseases (2). The binding affinities of protein complexes are related with sequence and structural features of residues at the interface. We have developed thermodynamic databases (3,4) for protein-protein, protein-nucleic acid and protein-carbohydrate complexes and derived several parameters including contact potentials, interaction energies and solvent accessibility to relate with binding affinity (5). Based on the insights gained from this analysis, we have developed computational tools for predicting the binding affinity of protein-protein (6), protein-DNA (7), protein-RNA (8) and protein-carbohydrate complexes. Further, we have elucidated the features important for the binding affinity change upon mutation and developed algorithms for predicting the change in binding affinity (9,10). These resources help to relate the binding affinity with disease-causing mutations, and designing therapeutic strategies for diseases.

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An HPC–AI framework for mapping druggable conformational landscapes of cancer proteins using Integrated Computing Environment (ICE)

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Protein conformational dynamics play a central role in cancer progression and drug response, yet capturing these transient states remains a major challenge in structure-based drug discovery. Recent advances in high-performance computing and AI-enabled automation provide new opportunities to efficiently explore biologically relevant conformational landscapes of cancer-associated proteins. We present an automated adaptive sampling molecular dynamics framework that accelerates the discovery of functionally important protein states relevant to drug binding. The workflow integrates large-scale molecular dynamics simulations with Markov state modeling–based analysis to iteratively identify metastable conformations and guide subsequent simulations toward underexplored regions of the conformational space. This closed-loop strategy enhances sampling efficiency while minimizing computational redundancy. The framework is deployed within an Integrated Computing Environment (ICE), enabling scalable execution, reproducibility, and seamless integration of simulation and analysis components. By combining AI-guided adaptive sampling with an automated HPC environment, this approach reduces time-to-insight and enables efficient mapping of druggable conformational states, supporting data-driven therapeutic discovery in computational oncology.



Invited speakers

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Specificity and promiscuity of phosphoinositide lipid interactions with GPCRs

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G protein-coupled receptors (GPCRs) are lipid-dependent membrane receptors that serve as important cell signaling hubs. Phosphoinositide (PIP) lipids represent an important class of anionic lipids that play vital roles in cellular function and signaling. PIPs have been reported to modulate GPCR function, although specificity and molecular details of the interactions are still not clear. In this work, we analyzed the specificity of the PIP lipid interactions with two GPCRs using coarse grain molecular dynamics simulations. Due to the importance of PIP lipids in receptor activation, we have considered the active state of the receptor, bound to its cognate G-protein (Gs or Gi). Our results show that four anionic lipid sites are present at the receptor surface, although the relative populations are dependent on the lipid type. In the case of the serotonin_{1A} receptor, a neuronal GPCR, phosphatidylinositol 4 monophosphate (PIP1) lipids exhibit highest interaction that is located at a charged cleft formed by transmembrane helices VI and VII. We observed electrostatic interactions at a cluster of charged residues (Arg341, Lys342, Lys345) and hydrophobic and aromatic interactions at residue Ile349 and Tyr402. In contrast, phosphatidylinositol 4,5 bisphosphate (PIP2) and phosphatidylinositol 3,4,5-trisphosphate (PIP3) lipids interact less. A reversed trend is observed in the case of the glucagon-like peptide-1 receptor (GLP-1R), an immune-metabulatory receptor, where PIP2 and PIP3 interactions dominate. Our work constitutes an important step to analyze molecular signatures of phosphoinositide lipid-GPCR interactions in the overall context of diverse roles of phosphoinositides in cellular function and signaling

Integrating biomolecular simulation with artificial intelligence

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In this seminar I will present some of recent investigations of my research group where we leverage machine learning and biomolecular simulation to address some of the projects related to structural biophysics. In particular, we will show how Generative Transformer models can be used to explore the role of GPT in decoding the molecular grammar of proteins undergoing liquid-liquid phase separation (LLPS). By training on curated datasets of LLPS-prone and resistant proteins, GPT models learn distinguishing sequence features and generate de novo peptides with phase-separating potential. In second part, we will show our recent effort in using machine translation in designing binder protein that would bind with intrinsically disordered regime and possible affinity prediction.

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Characterizing the transmembrane permeability of small molecules through human Aquaporin-10 using a “computational microscope”

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Macroscopic properties of biological matter are manifestation to their detailed atomic structure and dynamics. Due to limitation of spatial and temporal resolution, even most advanced experimental techniques are insufficient to characterize the biomolecular dynamics at nanoscale. To bridge this gap between experimental measurables and molecular level understanding of nanobiosystems, state-of-the-art classical molecular dynamics (MD) simulations method has immersed as reliable tool since past three decades. Building on the rapid advancements on computer hardware and simulation algorithms, at the age of exascale computing, MD simulations have been particularly successful in addressing wide range of research problems across structural, molecular and synthetic biology. Our research group at IIT Hyderabad is harnessing atomistic MD simulations to engineer synthetic nanobiosystems with prospective applications in therapeutics, desalination, sensing and sequencing.

In this talk, I will discuss results from an ongoing MD simulation study to characterize the structure and dynamics of human aquaporin 10 (hAQP10) channels embedded in lipid bilayer membranes. hAQP10 plays a critical role in glycerol metabolism and lipid homeostasis. This protein channel is implicated in several pathological conditions, including obesity and chronic dermatological condition pompholyx. A molecular-level understanding of the cohabitation and transport of water and glycerol molecules through the nanometer-sized pores of the hAQP10 channel is expected to enable rational design of therapeutics against these diseases.

Using equilibrium and advanced sampling MD simulations, we examine how the water and glycerol molecules diffuse amicably through the hAQP10 channel without any external bias. The residence time of glycerol in the channel varies from several nanoseconds to a couple of microseconds. Using replica exchange umbrella sampling (REUS) simulations, we computed the free energy profile of glycerol permeation along the channel axis. The estimated single-channel glycerol permeability, calculated using the modified inhomogeneous solubility–diffusion model, turns out to be 1.2×10^{-15} cm³/s, open to experimental validation.

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A modular and generalizable computational framework for mechanistic prediction of resistance mutations in infectious diseases

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Drug resistance remains a major challenge in infectious disease treatment, necessitating predictive frameworks that connect molecular mechanisms with clinical outcomes. Here, we present a generalized multiscale computational framework to decode resistance mutations in pathogen protein–drug or protein–antibody interfaces. Using the SARS-CoV-2 spike (S) protein and the monoclonal antibody sotrovimab as a model system, the framework integrates interface protein redesign, machine learning, hybrid quantum mechanics/molecular mechanics (QM/MM), all-atom and coarse-grained multiscale molecular dynamics (MD) simulations, and clinical correlation analyses. Starting with interface-focused redesign, the approach maps key interaction residues and identifies plausible escape mutations. Machine learning models leverage structural, sequence, energetics, and binding-affinity features to prioritize resistance-prone mutants. Hybrid QM/MM calculations further probe mutational hotspots as quantum regions to evaluate their impact on complex stability. Multiscale molecular simulations capture conformational dynamics, stability, and glycan-mediated effects, providing dynamic validation of static predictions. Predicted resistance mutations are assessed against clinical sequencing datasets and patient-derived evidence. Two residues—E340 at the S–sotrovimab interface and Y508 distal from it—exhibit strong concordance with clinically observed resistance signatures. Machine learning–derived new S-protein sequence variants showing enhanced or reduced sotrovimab affinity were structurally validated using AlphaFold. Overall, this modular and robust framework effectively identifies resistance mutations and aligns them with clinical insights. Beyond SARS-CoV-2, it offers broad translational potential for predicting emerging resistance in pathogens associated with urinary tract infections, tuberculosis, and fungal diseases, supporting rational design of next-generation therapeutics.

Keywords: *Resistance prediction, multiscale simulations, machine learning, interface redesign, pathogen evolution*

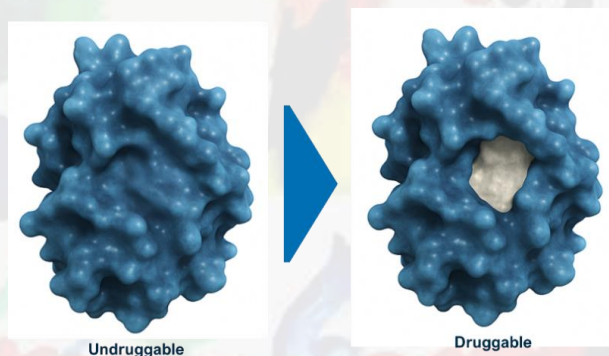
Making “undruggable” targets “druggable”

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Only one-fifth of the human proteome has been successfully modulated by small molecule drugs, leaving critical disease-associated proteins inaccessible. The key bottleneck is often the lack of an obvious, well-defined orthosteric binding site. This presentation addresses this major challenge by detailing a strategy rooted in computational detection of cryptic and transient binding pockets. A compelling case study on the E2 enzyme Cdc34, a difficult target where traditional screening failed. Through molecular dynamic simulation and advanced *in-silico* analysis, we successfully mapped hidden allosteric sites, demonstrating a path toward therapeutic intervention. This work highlights how Sygnature Discovery's integrated Computational Chemistry platform and secure HPC infrastructure are being used to break traditional druggability limits and expand the reach of small molecule programs.



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IN STRUCTURAL BIOPHYSICS

Design thinking for structural biologists: translating research skills to industry impact

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Structural biology delivers atomic-level insights that underpin modern drug discovery, biologics development, and diagnostic innovation. Yet, the transition from research outputs to industry-ready solutions often encounters challenges related to usability, scalability, and stakeholder alignment. Addressing these gaps requires an approach that integrates scientific rigor with human-centered design principles. Design Thinking offers such a framework. Its five stages, viz. Empathize, Define, Ideate, Prototype, and Test enable structured problem-solving that prioritizes end-user needs while fostering creativity and iterative development. These principles resonate strongly with the hypothesis-driven methodologies familiar to structural biologists, creating a natural synergy between scientific inquiry and design-led innovation.

Applications of this approach are evident across the life sciences industry. AI-driven drug design platforms leverage structural data while incorporating user-centric interfaces to enhance adoption. Cryo-EM software solutions have evolved through iterative prototyping to simplify workflows for non-expert users. Diagnostic tools have been refined through empathy-driven design to ensure accessibility in diverse clinical settings.

By aligning structural biology expertise with Design Thinking, research outputs can be transformed into impactful solutions that address real-world challenges in healthcare and pharmaceutical development. This integration not only accelerates innovation but also ensures that scientific advances translate effectively into outcomes that matter for patients and stakeholders.

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Cryo-EM structural determination of the Ysa ATPase from *Yersinia enterocolitica* reveals novel dodecameric and tetradecameric assemblies

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Yersinia enterocolitica is a Gram-negative bacterial pathogen responsible for gastrointestinal and systemic infections. Like many other Gram-negative pathogens, including *Salmonella*, *Shigella*, and *Pseudomonas*, its virulence relies on the Type III Secretion System (T3SS), a specialized nanomachine that delivers bacterial effector proteins directly into host cells. Central to T3SS function is a cytosolic ATPase that provides the energy required for substrate recognition, unfolding, and secretion. YsaN, the ATPase of the Ysa T3SS in *Y. enterocolitica*, is essential for secretion; however, its higher-order assembly and structural organization have remained largely unexplored. Our analysis identifies two distinct populations: a dodecameric and a tetradecameric complex, both exhibiting pronounced asymmetry. This is the first structural demonstration of such high-order oligomerization for a T3SS ATPase, highlighting an unexpected level of architectural complexity.

Detailed structural analysis suggests that these oligomeric states represent distinct functional assemblies. A conserved disulfide bond contributes to maintaining the integrity and stability of the oligomeric architecture. Structure-guided ATP docking reveals conserved nucleotide-binding sites supporting a coordinated ATP-binding mechanism within the oligomer, also supported by isothermal titration calorimetry (ITC) results. The structures reveal a well-defined central tunnel, consistent with a role in substrate engagement and unfolding during secretion.

In conclusion, our work uncovers an unprecedented oligomeric architecture of YsaN, establishing structural heterogeneity and higher-order assembly as key features of this T3SS ATPase. These findings provide a new structural framework for understanding energy transduction in the T3SS and significantly advance the current knowledge of secretion ATPase organization in Gram-negative bacterial pathogens.

Breaking the backbone of survival: targeting SufD–SufU in Iron–Sulfur cluster biogenesis of *Mycobacterium Tuberculosis*

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The global resurgence of tuberculosis, the emergence of multidrug-resistant *Mycobacterium tuberculosis* (Mtb), and the limited success of existing vaccine strategies collectively highlight the urgent need for therapeutics that target bacterial persistence beyond conventional drug targets. *Mtb* survival during oxidative stress and dormancy relies heavily on iron–sulfur (Fe–S) cluster biogenesis, which sustains respiration, metabolism, and redox balance. In *Mtb*, this process is mediated predominantly by the stress-responsive SUF pathway. Here, we identify the **SufD–SufU complex** as a critical and underexplored vulnerability within Fe–S cluster assembly. By integrating genome-wide essentiality data with structural and chemical biology approaches, our study highlights the role of Fe–S clusters in supporting Fe–S homeostasis under hostile host conditions. Structural insights into SufD–SufU interactions provide a framework for structure-based inhibitor design aimed at disrupting sulfur trafficking and Fe–S cluster formation. Inhibition of this pathway is expected to amplify intracellular oxidative stress and synergize with ROS-inducing antitubercular drugs, providing a promising strategy for targeting persistent and drug-resistant Mtb and reducing disease relapse.

STING agonists stung cyclic dinucleotide phosphodiesterase of *Mycobacterium Tuberculosis*

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Mycobacterium tuberculosis's (Mtb) specific cyclic dinucleotide phosphodiesterase, CdnP, serves dual roles: (A) it dampens host immunity by hydrolyzing Mtb's cyclic dinucleotide (CDN) c-di-3',3'-AMP and host-generated CDN 2',3'-c-GAMP, both known to trigger host immunity through Stimulator of Interferon Genes (STING) protein activation, and (B) it maintains 3',3'-c-d-AMP homeostasis crucial for Mtb's infection and pathogenesis. STING agonists are promising candidates for vaccine adjuvants and anti-tumor immune stimulants. Host lysed Mtb cells release CdnP in macrophages of the host and enhances pathogenesis of Mtb by degrading CDNs. Targeting CdnP could inhibit Mtb's infection and pathogenesis while enhancing host immunity. Sugar-modified mimetics of 2',3'-cGAMP containing arabinose and xylose sugar derivatives (23-(R/A/X)-cGAMP: R-Ribose, A-Arabinose & X-Xylose) have been reported as very potent STING agonists which are resistant to hydrolysis by the host's phosphodiesterase. We assumed that 23-(R/A/X)-cGAMP analogs could inhibit CdnP. Here, we report that 23-(R/A/X)-cGAMP analogs bind CdnP with a micromolar binding affinity. In addition, enzyme inhibition kinetic studies revealed these 2',3'-cGAMP mimetics exhibited micromolar inhibition constant (K_i) against CdnP. The co-crystal structure demonstrates that the 2',3'-cGAMP mimetic, 23-(R/A)-cGAMP, binds to the catalytic pocket with unusual 'horseshoe' conformation. In this conformation, the 23-(R/A)-cGAMP phosphodiester bond is inaccessible for cleavage, and it is nearly 10 Å away from the catalytic cation. Altogether, our study explores the design and characterization of CDN analogs inducing anti-mycobacterial activity and host immunity, paving the way for developing or repurposing more potent CdnP inhibitors with STING agonistic activities.

Understanding pyrethroid recognition and resistance through the structures of anopheles chemosensory proteins

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Mosquito-borne diseases are among the deadliest threats to global health, causing millions of deaths each year and imposing severe economic and social burdens. The emergence of multi-drug-resistant pathogen strains further complicates disease management, underscoring the urgent need for new therapeutic interventions. Effective control of mosquito populations remains a critical strategy in the fight against malaria. Although recent advances in genomics have enabled the decoding of mosquito genomes and the identification of key proteins involved in essential biological processes, the structures of most mosquito proteins remain largely unexplored. Without detailed structural insights into these molecular targets, the development of innovative mosquito-control strategies remains challenging. Therefore, expanding our understanding of mosquito protein structural biology is crucial for guiding new interventions.

Insects rely on a highly specialized olfactory system composed of chemosensory proteins (CSPs), odorant-binding proteins (OBPs), and their associated receptors. These proteins detect and bind organic volatiles, enabling the processing and transmission of chemical signals to the nervous system. In mosquitoes, CSPs and OBPs are central to host recognition, behavioural modulation, and ecological adaptation. Yet, despite their importance, our understanding of how these proteins discriminate among diverse chemical volatiles, the molecular mechanisms governing ligand binding, and the subsequent relay of signals to sensory pathways remains limited. Notably, a recent study by Sharma *et al.* identified SAP2 from *Anopheles gambiae* a prominent CSP family member as a critical factor in insecticide resistance, though the mechanistic basis for its role is still not fully understood.

Recently, we conducted a comprehensive investigation of CSPs and OBPs from multiple *Anopheles* species using an integrated structural, computational, and biochemical approach. Our findings are highly exciting, as they reveal detailed insights into how these proteins bind to volatile molecules, undergo conformational changes, and facilitate downstream signal transmission. This integrated analysis has allowed us to pinpoint key residues and structural motifs involved in ligand recognition and specificity, shedding light on the molecular logic of mosquito olfaction.

In my presentation, I will elaborate on the biochemical and structural characteristics of several CSPs and their homologues, including the D7 protein, across three *Anopheles* species. I will further discuss how these proteins may influence vector competence, govern host-seeking and invasion behaviours, and contribute to emerging patterns of insecticide resistance.

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Molecular switches that regulate actin cytoskeleton nucleation

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The actin cytoskeleton is a dynamic filamentous network essential for processes such as cell migration, environmental sensing, shape maintenance, and vesicular trafficking. Actin filament assembly begins with a rate-limiting nucleation step, which is strongly accelerated by Arp2/3 complex. While Arp2/3 complex is indispensable for branched actin network formation, it remains inactive without stimulation by nucleation-promoting factors (NPFs). Precise spatiotemporal control of actin nucleation is therefore achieved through regulation of these NPFs.

The Wiskott–Aldrich Syndrome Protein and SCAR Homolog (WASH) is the principal endosomal NPF, driving the recycling of internalized proteins back to the cell surface to maintain membrane protein homeostasis and regulate signalling. WASH functions as part of the large, multi-subunit WASH Regulatory Complex (SHRC). Despite its central role, the mechanistic regulation of WASH within SHRC has remained poorly understood due to the lack of high-resolution structural information on the complex or its subunits.

To address this, we performed high-resolution electron cryo-microscopy (cryo-EM) structural studies to resolve the architecture and conformational dynamics of SHRC. We obtained structures of SHRC in multiple states, revealing that WASH intrinsically toggles between nucleation-competent and inactive forms. This intrinsic basal activity distinguishes WASH from canonical NPFs with complex regulatory control. Biochemical assays validated these structural findings, while molecular dynamics simulations provided mechanistic and thermodynamic insights into SHRC inhibition and activation. Our results establish a structural and mechanistic framework for SHRC regulation, setting the stage for future studies of disease associated mutations and additional cellular factors that modulate WASH activity.

Decoding the molecular basis of catalytic mechanism and allosteric regulation of Glutamate Dehydrogenases (Gdhs)

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Glutamate dehydrogenase (GDH) is a key enzyme that connects carbon and nitrogen metabolism in all living organisms. Despite of studies on this enzyme for past several decades, Capturing different structural intermediates for gaining insights into the reaction mechanism and kinetic properties, and allosteric regulations in GDHs have been very challenging. Extensive structural and biochemical studies on GDHs are essential. We have determined the first crystal structure of a fungal GDH. We report the structural basis of the unique properties related to the catalytic mechanism and allosteric behavior of fungal GDHs. The high resolution structures of GDHs from *Aspergillus niger*, *Aspergillus terreus* and *Saccharomyces cerevisiae*, as its apo form, complexed with substrate (a-ketoglutarate), coenzyme (NADPH) and reaction intermediates (a-aminoglutarate and 2-amino-2-hydroxyglutarate) provide molecular details to understand the catalytic mechanism and coenzyme specificity of the enzyme. Our study revealed important structural features essential for positioning a-ketoglutarate, NADPH, ammonium ion and the transient intermediates in the GDH active site. Based on our data the steps of the catalytic reaction mechanism of GDH have been deciphered. Cryo-electron microscopy (cryo-EM) structures of GDHs with different conformational states have also been captured. The hexameric cryo-EM and crystal structures of the fungal GDHs have revealed some of the unique conformations of this enzyme. Our biochemical and structural data identified domain motion and molecular network to be responsible for the allosteric regulation in hexameric GDHs.

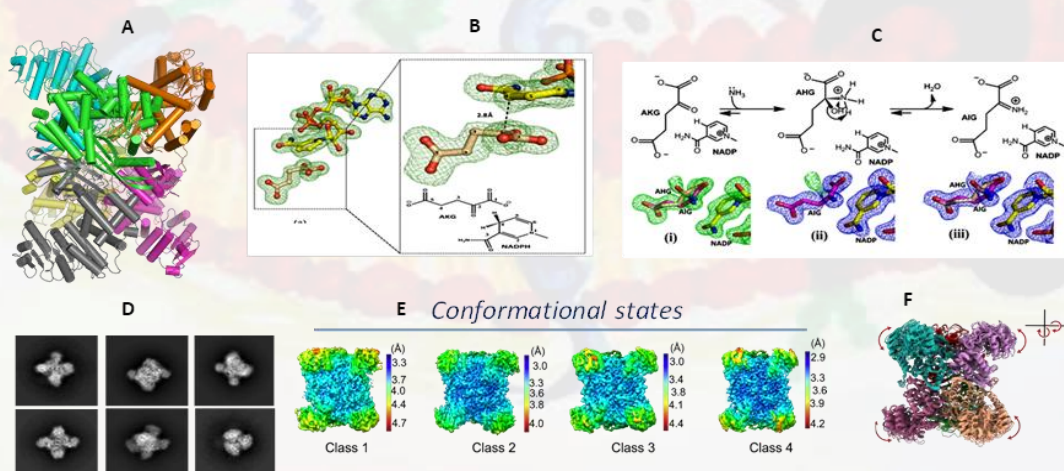


Figure 1. Structures of fungal GDH explain functional properties of the enzyme. **A.** Hexameric structure of GDH. **B.** s-A weighted F_o-F_c omit electron density maps (3.0s level) of a-ketoglutarate (AKG) (light brown carbon) and NADPH (yellow carbon) showing the close proximity of substrate and coenzyme for hydride transfer. **C.** Electron density maps ($2F_o-F_c$: blue and F_o-F_c : green) revealed presence of reaction intermediates in the active site of GDH. **D.** Conformational states of GDH after 2D classification of Cryo-EM map. **E.** 3D density maps from Cryo-EM data indicating conformational states of GDH. **F.** Conformational dynamics of GDH structures associated to allosteric regulation of the enzyme.

Keywords: Glutamate dehydrogenase (GDH), Allosteric regulation, Reaction intermediate

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Cold denaturation of proteins in the presence of additives: is cooling a reflective image of heating?

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Proteins undergo denaturation under both extremes of temperature. Cold-induced denaturation is driven by negative enthalpy and follows a different conformational pathway from the corresponding heat-induced denaturation. This implies that additives may produce differential effects on the cold and heat denaturation process of a protein. This phenomenon is tested with two types of additives: polyols with varying numbers of hydroxyl groups and surfactants with different charges and alkyl chain length. The analysis of basic thermodynamic parameters and structural changes indicates that polyols stabilize protein against cold denaturation, whereas they destabilize the protein during heat denaturation. The extent of this effect increases with an increase in hydrophobic interaction between the protein and the polyols. In the case of surfactants, the addition of ionic surfactants destabilizes the protein during both heat and cold denaturation. However, non-ionic surfactants, at their micellar concentrations, marginally increase the stability of protein against heat denaturation. During cold denaturation, the micelles induce partial unfolding but resist complete denaturation, a unique behavior observed only in the presence of non-ionic surfactants during the cold denaturation of a protein.

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Structure-guided prediction of protein expression

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Predicting protein expression from amino acid sequences is important for protein engineering and functional genomics, but current approaches lack scalability and structure integration. We have developed a modular pipeline that predicts protein expression using sequence and structure features combined with machine learning. The system extracts physicochemical and compositional properties from sequences, estimates structural features such as solvent accessibility and secondary structure, and trains LightGBM-based models on DepMap expression data. It supports parallel feature extraction, caching, and optional use of ESMFold and ESM2 embeddings. This framework is unique because it links real expression data with structure-aware features in a reproducible and scalable way. The tool provides a practical foundation for predicting protein expression and guiding sequence optimization in experimental design.

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Structural biology for accelerated drug discovery

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Structural Biology has been at the forefront of drug discovery elucidating three dimensional structures of proteins with small molecules and/or complexes with other proteins, nucleic acids, and carbohydrates. The detailed atomic level information provided by these structures has enabled rational structure-based drug design, target identification and therapeutic lead generation in the early pre-clinical discovery stages. Combining structural biology with multiple disciplines, such as, target validation, biophysical characterization, assay biology, computational chemistry, and medicinal chemistry, an Integrated Drug Discovery (IDD) workflow can be created that accelerates hit-to-lead and lead optimization phases. X-ray crystallography, cryo-electron microscopy (cryo-EM), and NMR spectroscopy allow visualization of binding sites, conformational changes, and ligand interactions at atomic resolution. These insights aid fragment-based drug discovery (FBDD), virtual screening, and iterative optimization cycles, reducing attrition in later stages, improving the success rates and reducing the timelines and cost of drug development. Sai Life Sciences is a Contract Research, Development and Research Organization (CRDMO) that offers end-to-end IDD services, designed to accelerate the journey from target identification and validation to IND filing. Our IDD model brings together multiple disciplines under one roof, ensuring close cross-functional collaboration and faster decision-making cycles.

Semiflexibility speeds up the translocation of multiple polymers in nanopores

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The process of polymer translocation (PT) through narrow pores is essential for a number of biological systems. Here, we investigate an unforced PT of a single polymer and of multiple polymers through a pore in two dimensions using the Langevin-dynamics simulations using a coarse-grained bead-spring model. We aim to investigate under which conditions "stacking" multiple polymers inside a pore facilitates their cooperative PT. The results show that increasing the bending stiffness κ of a single polymer slows down the PT process by reducing the configurational freedom and inhibiting the tension propagation along the chain. In contrast, when two polymers occupy the pore simultaneously, their bending stiffness can *speed up* the PT. Two flexible polymers translocate faster than a single flexible chain due to "entropic pushing", while two semiflexible polymers increasingly prefer the same-side PT as their stiffness increases. In "mixed" situations of a simultaneous PT of a flexible and a semiflexible polymer, the semiflexible chain consistently translocates faster due to reduced entropic trapping and more efficient tension propagation. These findings demonstrate that bending stiffness, which typically limits the single-PT dynamics, can counterintuitively enhance the cooperative transport in the multiple-polymers scenarios. This can provide new insights into the peculiarities of the PT in a number of systems featuring biological channels or synthetic nanopores.

CONFERENCE ON EMERGING TRENDS
IN STRUCTURAL BIOPHYSICS

Harnessing deep neural networks and large-scale screening of chemical libraries to explore novel chemical space for JAK1 inhibitors

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Janus kinase 1 (JAK1) is a non-receptor tyrosine kinase central to cytokine-mediated signaling in autoimmune, inflammatory, and oncologic diseases. Current JAK inhibitors such as Tofacitinib and Ruxolitinib are constrained by their common pyrrolo[2,3-d]pyrimidine scaffold, which results in poor isoform selectivity, off-target binding, and associated toxicities. To overcome these challenges, we developed a scalable AI-driven virtual screening pipeline to identify chemically novel and selective JAK1 inhibitors.

Our deep neural network (DNN) model was trained on docking scores generated from an ensemble of eight distinct JAK1 crystal structures, capturing protein flexibility and yielding a broad activity landscape. Averaged docking scores were used to define actives and inactives, allowing the model to learn from a structurally diverse dataset.

Screening the ZINC database compounds resulted in ~13,000 candidates. Many showed diverse chemical scaffolds compared to known JAK1 inhibitors. Ensemble docking further validated their binding potential.

This work represents the first machine-learning framework trained on ensemble docking scores across multiple protein conformations, enabling billion-scale virtual screening to uncover novel scaffolds for selective JAK1 inhibition. The findings present a new paradigm for incorporating protein flexibility into AI-guided drug discovery, thereby expanding access to previously unexplored chemical space.

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Oral Presentations

CONFERENCE ON EMERGING TRENDS
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Proteins denature upon both increase and decrease in temperature. What drives the proteins to attain a disordered structure at low temperatures is an intriguing thermodynamic phenomenon. The most popular hypothesis states that at critically low temperatures, protein is forced to expose its hydrophobic core to water (Privalov, Biochem. & Mol. Bio, 1990) and its reduced solvation energy makes it to unfold. To study the role of this hydrophobic hydration during cold-induced denaturation, iron sulphur cluster unit (IscU) protein, which unfolds above the freezing point of water, was chosen. The point mutations replacing charged or polar residues with hydrophobic residues, E25V, D49V, and a double mutant E25V- D49V were produced. Their thermal stability against heat and cold was analysed in varying ionic strengths. The D49V and E25V-D49V increased the stability of IscU against both heat- and cold-induced denaturation, but the E25V showed only marginal stabilization against cold unfolding. The increase in ionic strength showed marginal stability in all the mutants compared to the wild-type. Furthermore, the point mutation to reverse the charge of the polar residues, E25K and D49K, and double mutants E25K/D49K and E25K/D49V were produced. Though both the mutants, E25K and D49K increased the stability of the protein against both heat and cold-induced denaturation, and increasing the salt concentration showed significant stabilization when compared to the wild type protein. However, the D49K showed a strong stabilization effect (up to 6 K) among all the mutants. The results suggest that the hydrophobic mutations stabilize the protein preferentially against the heat unfolding by reducing the flexibility of loops and/or removing electrostatic frustration, whereas the charge reversal mutations stabilize the protein against both heat and cold-induced denaturation might be by establishing new electrostatic interactions with other residues and reducing the favourable hydrophobic hydration. This study highlights the role of surface residues on the thermal stability of proteins at extreme temperatures, and understanding of which is crucial for many fields, such as protein engineering, cryopreservation, and thermal adaptation of organisms.

Keywords: Cold denaturation, IscU, Thermal unfolding

Structural rewiring of the mannose-binding pocket reveals pathoadaptive FimH mutations that enhance uroplakin-associated glycan recognition and uropathogenic adhesion

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Urinary tract infections (UTIs) constitute a major global health concern, with uropathogenic *Escherichia coli* (UPEC) serving as the predominant causative agent. The adhesive competence of UPEC is governed by the fimbrial lectin FimH, which mediates attachment to high-mannose glycans displayed on the urothelial uroplakin (UPK) complex. Although structural insights have delineated FimH interactions with independent glycans, its recognition of UPK1A-tethered glycans within the intact human UPK architecture remains unresolved. Moreover, the mutational logic by which the Mannose-Binding Pocket (MBP) undergoes structural rewiring to enhance glycan affinity and confer pathoadaptive advantage has not been systematically explored. In this study, multiscale structural modelling, interface- based protein redesign and mutational energy profiling were employed to elucidate FimH–UPK–glycan interactions and uncover affinity-modulating mutations. Using the porcine UPK structure as a scaffold, human UPK assemblies complexed with oligomannose-3 (OM3) and oligomannose-6 (OM6) were constructed and analyzed. Comparative redesign of the FimH MBP yielded 545 mutated-designs across human and porcine complexes, from which affinity- and stability-enhancing pathoadaptive mutations—such as N135S and Y137F (human) and D54G and D47V (porcine)—were identified in complexes with OM3 and OM6, respectively. Interaction energy decomposition revealed an increased density of interfacial hydrogen-bonding and stabilizing van der Waals contacts in these designs, concomitant with higher conformational rigidity as demonstrated by normal mode analyses and their dual role in reinforcing glycan-binding and MBP robustness. These findings delineate the structural- energetic basis for FimH’s evolutionary fine-tuning toward UPK-associated glycans, providing mechanistic insight into UPEC adhesion, and informing rational strategies for anti- adhesive therapeutic design.

Keywords: *Urinary tract infections, FimH adhesin, Pathoadaptive mutations.*

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Pancreatic cancer is a highly aggressive and often lethal malignancy with limited treatment options. Its late-stage diagnosis and resistance to conventional therapies make it a significant challenge in oncology. Immunotherapy, particularly cancer vaccines, has emerged as a promising avenue for treating pancreatic cancer. Multi-epitope vaccines, designed to target multiple epitopes derived from various antigens associated with pancreatic cancer, have gained attention as potential candidates for improving therapeutic outcomes. In this study, we have explored transcriptomics and protein expression databases to identify potential upregulated proteins in pancreatic cancer cells. After examining a total of 21,054 proteins from various databases, it was discovered that 143 proteins expressed differently in malignant and healthy cells. The CTL, HTL and BCE epitopes were predicted for the shortlisted proteins. 51,840 vaccine constructs were created by concatenating CTL, HTL, and B-cell epitopes in the respective sequences. The best 86 structures were selected from a set of 51,840 designs after they were analyzed for vaxijenicity, allergenicity, toxicity, and antigenicity scores. In further simulation of the immune response using constructs, it was found that 41417, 37961, and 40841 constructs could produce a strong immune response when injected. Further, it was found that construct 37961 showed stronger interaction and stability with TLR-9 as determined from the large-scale molecular dynamics simulations. Moreover, the 37961 construct has shown interactions with TLR-9 suggests its potential in inducing immune response. In addition, construct 37961 has shown 100% predicted solubility in the *E. coli* expression system. Overall, the study indicates the designed construct 37961 has the potential to induce an anti-tumor immune response and long-standing protection pending further studies.

Keywords: *Multi-epitope vaccines, pancreatic cancer, BCE epitopes.*

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DNA tetrapods, which combine DNA's programmable binding with a tetrapod's symmetric, porous architecture and uniform force-distribution property, are promising building blocks for structural DNA nanotechnology. Studying their self-assembly reveals the principles that govern the formation of complex structures and functional devices. A coarse-grained bead-spring model was employed for the DNA tetrapod, with double-stranded arms represented by inert beads and single-stranded arm terminals represented by reactive beads. Self-assembly was induced by lowering the system temperature, which reduces pairwise interaction energy and promotes bond formation. Analysis of the radial distribution function, structure factor, and dynamic slowdown reveals a liquid-to-gel phase transition. We analysed the morphology of the equilibrium gels formed by self-assembly. Further study of these systems will improve understanding of the physics of self-assembly and provide a basis for controlling the morphology and mechanical properties of equilibrium gels by tuning the properties of constituent DNA tetrapods.

Keywords: *DNA tetrapods, self-assembly, structural DNA nanotechnology.*

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In medical practice, brain tumours are still difficult to identify early and diagnose accurately, which frequently results in postponed therapies and unfavourable patient outcomes. Promising approaches to improve brain tumour diagnosis, classification, and prognosis are provided by recent developments in machine learning (ML) technologies. With an emphasis on imaging data like MRI and CT scans, this study explores the use of machine learning approaches in brain tumour detection and management. The effectiveness of several machine learning methods, such as random forests, support vector machines, and deep learning, in automating tumour recognition, segmentation, and classification is investigated. Based on the evaluation, this model accurately categorized 1 case as tumor and 0 cases as non-tumor. The model achieved an average tumor probability of 99.74%, highlighting its confidence in predicting. This model reaches an accuracy of 80%, with a precision of 85%, a recall of 75%, and an F1 score of 79.5%. Through the analysis of imaging features and clinical data, the study also assesses the ability of machine learning algorithms to forecast tumour growth and patient survival outcomes. Initial findings show that ML models can perform better than conventional.

Keywords: *Image segmentation, CNN, Image classification.*

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Alzheimer's disease (AD), a progressive neurodegenerative disorder, is driven by complex molecular processes and exacerbated by diverse risk factors. While amyloid- β plaques and tau tangles have traditionally been considered hallmarks of AD, recent studies suggest that ferroptosis plays a central role in AD pathogenesis. Ferroptosis is an iron-dependent regulated cell death marked by lipid peroxidation and oxidative stress. In this study, we established a robust molecular link between AD and ferroptosis using transcriptomic profiling and pathway enrichment analysis and evaluated how major AD risk factors such as aging, chronic alcohol use, alcohol-nicotine co-abuse, short-term memory dysfunction, comorbidities and family history influence ferroptosis-associated gene expression patterns.

Through differential gene expression analysis of multiple GEO datasets, we identified consistent enrichment of ferroptosis-related pathways across all risk factor profiles. Seven key ferroptosis associated genes (CYBB, FERMT1, BAX, SOD1, ACSL4, TP53 and FTH1) were commonly dysregulated. Machine learning models (LASSO and Random Forest) followed by neural network-based diagnostics prioritized these genes as highly predictive biomarkers ($AUC \geq 0.80$). Genem-miRNA interaction networks revealed that these genes are regulated by several hub miRNAs, including hsa-miR-34a (TP53, SOD1, BAX), hsa-miR-34b and hsa-miR-34c (TP53, FERMT1), hsa-miR-125a-5p (FERMT1, TP53) and hsa-miR-20a-5p (ACSL4, BAX), which are associated with oxidative stress, neuroinflammation, synaptic dysfunction and immune regulation. Immune infiltration analysis demonstrated upregulation of macrophages, monocytes and T cells in AD, suggesting that ferroptotic damage may trigger neuroinflammatory cascades and create a vicious cycle of ferroptosis-driven immune activation. Finally, compounds such as glutathione and alpha-tocopherol were identified as potential therapeutic agents for targeting ferroptosis via drug repurposing. By integrating ferroptosis biology with multi-risk factor analysis, gene-miRNA networks and immune profiling, this study provides a comprehensive framework for early AD diagnosis and highlights new avenues for precision therapeutics targeting ferroptosis-driven neurodegeneration.

Keywords: *Alzheimer's disease (AD), Ferroptosis, Machine Learning*

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Network motifs, such as feedback loops, feed-forward loops (FFLs) and their higher-order arrangements known as hypermotifs, serve as fundamental building blocks that orchestrate cellular decision-making networks in cancer. In the context of Epithelial-Mesenchymal Transition (EMT) - a hallmark of cancer metastasis, these motifs coordinate intracellular and extracellular signals across diverse pathways including TGF β , Wnt- β -Catenin, NOTCH and several Receptor Tyrosine Kinases (RTKs). Central to this regulation are core transcription factors such as SNAIL, ZEB embedded within feed-forward and feedback loops mediated by both SMAD-dependent and SMAD-independent TGF β pathways. Their placement within specific motifs provides them with distinct dynamical properties that govern phenotypic transitions. This systems level study explores the role of motifs and hypermotifs observed in EMT regulation through dynamic modelling and analysis. The results illustrated the information processing role of major EMT transcription factor SNAIL along with its system level behaviour. SNAIL which is regulated in a coherent feed-forward loop motif exhibited noise filtering, signal integration, persistence detection, coincidence detection and temporal regulation during EM transition. Further analysis of the feedback loops involving p53 illustrates that these motifs drive multistable dynamics that underpin phenotypic plasticity and enable transitions among epithelial, hybrid, and mesenchymal states. These behaviours highlight that these regulators act as control points within the EMT decision-making landscape. These findings uncover the coordinated interplay among key regulators (such as p53, SNAIL, ZEB, MDM2, GLI, E-cadherin, N-cadherin, SMADs, miR-200, and miR-34), and their associated feed-forward and feedback circuits, establishing EMT controllers as structural targets that encode robustness, reversibility during metastatic progression. Targeting such decision-making regulators may offer a more effective route to modulate EMT. By characterizing the dynamic behaviour of EMT regulating motifs, this work identifies key molecular regulators whose systems-level roles make them promising candidates for structure guided interrogation and therapeutic intervention.

Keywords: EMT, Network Motifs, SNAIL

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Breast cancer remains a significant global health challenge due to its molecular heterogeneity and diverse clinical stages, which hinder accurate prognosis and effective treatment. The absence of specific therapeutic targets for certain subtypes and stages often limits the precision of current interventions, resulting in variable efficacy, drug resistance, and increased toxicity. There is an urgent need for integrative approaches to identify prognostic biomarkers and novel, less toxic treatment options. This study aimed to develop and apply an integrated framework combining machine learning and bioinformatics to identify prognostic biomarkers and potential natural-product-based therapeutic candidates for breast cancer, with a focus on clinically challenging cases such as the Basal subtype and Stage 4 disease. Matched mRNA expression and clinical data from The Cancer Genome Atlas were curated and analyzed for differentially expressed genes, which were validated using Gene Ontology and pathway enrichment analyses. Six ML algorithms (RF, SVM, KNN, Logistic Regression, XGBoost, and DNN) were used to build predictive models. The top-performing model's selected features underwent prognostic analysis. In focusing on the Basal subtype and Stage 4 breast cancer, common genes were identified and validated through literature review. Docking studies against a curated and ADMET-filtered database of natural compounds (NPASS) revealed potential candidates with favorable binding affinities to the identified targets. The proposed integrated framework effectively identified prognostic biomarkers and screened natural products with therapeutic potential for high-risk breast cancer subtypes and stages. This approach presents the possibility of developing more targeted, and less toxic therapeutic strategies.

Keywords: *Breast cancer, Differential expressed genes, Machine learning*

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β -glucosidase is the rate-limiting enzyme in the enzymatic degradation of cellulose for industrial bioethanol production, which catalyzes the conversion of cellobiose to glucose (1). The major challenges faced by β glucosidases in industrial setting include inhibition by its product glucose and maintaining its activity at high temperature (50-60°C) and low pH (4.5 – 5.5). Protein engineering is employed to improve properties of β glucosidase as per industrial requirement. Our group has extensively worked on increasing glucose tolerance of β -glucosidases by structure-based rational engineering and patented multiple glucose tolerant variants of β glucosidases (2,3). However, the scope of modulating pH optima of β -glucosidases towards acidic range remains unexplored. We have initiated the work on shifting pH optimum towards acidic side by engineering of AtGH1 (a GH1 β -glucosidase from *Acetivibrio thermocellus*,) having pH optima 5.5 with the rationale developed based on sequence and structural alignment with FoBgl (another GH1 β -glucosidase from *Fusarium oxysporum*) showing broader pH optima profile of 4.5-7.0. A mutant of AtGH1 was successfully obtained showing shift in pH optimum from 5.5 to 5.0 along with improved kinetic properties (3). To further understand the molecular mechanism of modulation of pH optima, high resolution structural information is essential. Therefore, we have utilized UnBG11 (another GH1 β -glucosidase from soil metagenome) as model system which is known to form robust crystals diffracting to atomic resolution. Similar mutations in UnBG11 showed shift in pH optimum from 6 to 5.5, which were subjected to crystallization (Figure. 1a). The crystals of UnBG11-mutants diffracted to atomic resolution of ~ 0.99 – 1.2 Å at PX-BL21 beamline, RRCAT, Indore, India (4). The high-resolution crystallographic data have been processed, and structures of the UnBG11-mutants have been solved. The refinement of these atomic resolution structures is in progress (Figure 1b). Hence, the structural information at the atomic resolution will help to decipher the molecular mechanism of shift in pH optima of UnBG11 as well as other GH1 family β -glucosidases.

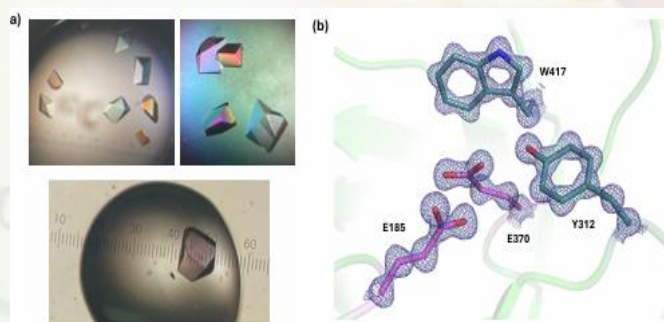


Figure 1. (a) Representative images of crystals of UnBG11-mutants, formed within 1-3 days using crystal seeds in mother liquor (0.056M monobasic sodium monohydrate, phosphate 1.344 M potassium phosphate dibasic, pH 8.2). The crystals ranged from ~ 0.5 mm to 1 mm in size. (b) The $2F_o - F_c$ map (blue mesh) at 1.0 Å resolution contoured at 2.5σ level showing the fully satisfied electron density around active site residues. Catalytic glutamates are presented as pink stick and other active site residues are represented as teal sticks.

Keywords: *GHI β -glucosidase, pH optima, atomic resolution crystal structure*

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The inhibitory effects of tamarind (*Tamarindus indica*) xyloglucan polysaccharides, locust bean gum, guar gum, and copra meal on α -glucosidase (*Saccharomyces cerevisiae*), as well as the inhibitory effects of xyloglucan on the chimeric enzyme [ManB-(GGGS)₃-XynB], were investigated. Xyloglucan showed no observable inhibitory action towards the chimeric enzyme under the tested conditions. Similarly, locust bean gum, guar gum, and copra meal exhibited no inhibitory activity towards α -glucosidase. In contrast, xyloglucan strongly inhibited α -glucosidase, and its inhibitory behaviour and kinetic parameters were subsequently analysed using Lineweaver–Burk plots. Biophysical analyses, including Fourier-transform infrared (FT-IR), ultraviolet-visible (UV-Vis), circular dichroism (CD), and synchronous fluorescence spectroscopy, were performed to examine structural and micro-environmental changes in α -glucosidase upon interaction with xyloglucan. Furthermore, steady-state fluorescence and molecular docking were employed to elucidate the molecular basis of inhibition and binding interactions. Xyloglucan polysaccharides showed significant inhibitory activity toward α -glucosidase, with an IC_{50} of 350 μ g/ml and exhibited a mixed-type inhibition mechanism ($K_m = 1.16$ mM; $V_{max} = 196.5$ mM/min). UV absorption spectra showed enhancement in the absorption intensity in the presence of xyloglucan polysaccharides, indicating enzyme-polysaccharide interaction. FT-IR spectra confirmed that complex formation between α -glucosidase and polysaccharide did not alter the characteristic functional groups of the enzyme. CD spectral analysis further demonstrated that xyloglucan induced conformational changes in the secondary structure of α -glucosidase (decreased α -helix, β -sheet content and an increase in random coil content). Intrinsic fluorescence of α -glucosidase was efficiently quenched by xyloglucan polysaccharides through a static mechanism, as reflected by a decrease in the quenching constant from 0.4484 to 0.3792 L mg⁻¹ with increasing temperature. Synchronous fluorescence indicated alterations in the microenvironment surrounding tryptophan and tyrosine residues, while ANS fluorescence suggested decreased exposure of hydrophobic regions in the presence of xyloglucan polysaccharides. Molecular docking revealed strong binding affinity between xyloglucan and α -glucosidase, with a binding energy of -9.135 kcal/mol, primarily stabilised by hydrogen bonding (Asp325, Ser304, Pro320, Arg315, Thr310, Pro312, Glu332, Asp242, Lys156, Ser241, Tyr158) and van der Waals interactions. Overall, these findings highlight that xyloglucan polysaccharides can be promising natural α -glucosidase inhibitors and can be used in the management of diabetes.

Keywords: α -glucosidase, Tamarind xyloglucan, Spectroscopy

Crystallographic studies of Glucose-6-Phosphate Dehydrogenase for designing novel inhibitors against Leishmaniasis

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The enzyme glucose-6-phosphate dehydrogenase (G6PDH) plays a critical role in trypanosomatid parasites, including *Leishmania donovani*, by generating NADPH to combat oxidative stress. In this study, LdG6PDH was successfully cloned, expressed, and purified to homogeneity using affinity chromatography. Size exclusion chromatography displayed a single peak corresponding to approximately 132 kDa, indicating that the enzyme exists as a dimer. Enzymatic characterization showed that the purified enzyme exhibits optimal catalytic activity at a pH close to physiological conditions. Kinetic analyses demonstrated that LdG6PDH has a higher affinity for its cofactor NADP than for the substrate glucose-6-phosphate (G6P). Enzyme inhibition assays identified ellagic acid (EA) and epigallocatechin gallate (EGCG) as potent inhibitors of LdG6PDH, both acting through a mixed-type inhibition mechanism with respect to the substrate. Secondary structure analysis using circular dichroism (CD) spectroscopy deciphered that the enzyme is predominantly α -helical and remains structurally stable over a range of pH values. The crystal structure of LdG6PDH was determined using the hanging-drop vapor diffusion method at a resolution of 2.4 Å, with crystals belonging to the monoclinic space group C121. The asymmetric unit contains two protomers, with a Matthews coefficient (VM) of 2.45 Å³ Da⁻¹. Ligand bound crystal structures of LdG6PDH were obtained by soaking with G6P and glucosamine-5-phosphate (GLP), which diffracted to resolutions of 2.3 and 2.25 Å, respectively. These complex structures revealed distinct ligand-binding modes compared to previously reported G6PDH enzymes. Structural analysis delineated that the N-terminal domain is unique to *Leishmania* G6PDH when compared with the human enzyme. This comprehensive structural and biochemical characterization of LdG6PDH provides valuable insights into its function and inhibition, laying a strong foundation for the rational design of drugs targeting this enzyme for the treatment of leishmaniasis.

Keywords: *Leishmania donovani*, glucose-6-phosphate dehydrogenase, crystal structure.

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Protein aggregation and its mechanisms remain an enigma for protein biochemists and are not yet completely understood. One of the primary causes of protein aggregation is the loss of cell-quality control mechanisms due to ageing; as a result, most proteinopathies are associated with ageing. The aggregation of the crystallin protein is primarily responsible for cataract. It occurs when lens aggregation increases light scattering in the eyes, leading to vision loss. These aggregations arise for several reasons, including covalent modifications in proteins that accumulate throughout a person's lifetime, forming a heavy burden in ageing demography. In the current work, we have chosen wild-type Gamma d-crystallin protein (PDB ID: 1HK0) and two specific mutants (name the mutants), which, upon analysis, were found to be associated with severe pathology among various populations. We used AlphaFold to generate mutant models and performed subsequent quality analysis and model refinement using Galaxy TBM. Our final validation of the models was performed using the ProSA and PROCHECK servers, resulting in an acceptable Ramachandran plot. For our molecular docking, we are using naturally occurring small molecules from the INFLAMNAT Database, a subset of the COCONUT database. We are performing toxicity screening and PB/PK simulations for the molecules, followed by molecular docking and molecular simulations. We aim to translate in-silico research into more clinically translational options, increasing the use of natural product-based therapeutics for cataract.

Keywords: cataract, gamma-D-crystallin, inflammation, protein aggregation, alphafold

Designing novel peptide inhibitors against Gamma-D-Crystallin: an *In-Silico* peptide-based therapeutic approach for crystallin aggregation

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Cataract is the clouding of the eye's usually clear lens, which causes blurred or dim vision. This cloudiness happens when proteins in the lens clump together. The most common cause of cataract is ageing, but other factors include eye injury, diabetes, genetics, and prolonged use of certain medications like steroids. Gamma-crystallin plays an essential role in reducing lens transparency with age. They are also subjected to various destabilising mutations and other chemical assaults, which lead them to lose their native structure and form aggregates. This process leads to protein precipitation that causes the cloudy appearance. Treatment of cataract involves surgical interventions and post-operative care. Novel peptides are an interesting strategy with high therapeutic potential for proteinopathies. In the current work, we have chosen two mutants, R80S and W43Q, which were found to be deleterious among many gamma-crystallin protein mutants when analysed using Sorting Tolerant from Intolerant (SIFT) and FATHMM. The mutant models were generated by alphafold. These models were analysed with MOLPROBITY, and the one with the highest MOLPROBITY score, low clash scores, and poor rotamers was chosen for further refinement using the Galaxy Refine 2 server. We reanalysed the models using MolProbity and compared clash scores, galaxy energy, and the number of poor rotamers. The secondary structure analysis was done on the basis of the Ramachandran plot. We are currently designing novel small peptides using RF Diffusion methods. The goal is to perform protein-peptide docking followed by molecular simulations and to analyse their anti-aggregation potential. Selected peptides identified from our *in-silico* work will be synthesised for further validation using gamma-D-crystallin protein-mediated aggregation.

Keywords: *Gamma-D-crystallin, peptides, proteinopathies, RF Diffusion, mutation*

Structure-based identification of selective inhibitors targeting *Leishmania Donovanii* Xaa-Pro dipeptidase

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Xaa-Pro dipeptidase (XPD) is a metalloprotease that cleaves dipeptides ending with proline, a residue that restricts backbone flexibility. Although this enzyme plays an important role in parasitic systems, its structural characteristics in protozoan pathogens remain insufficiently explored. In this study, an integrated experimental and computational strategy was employed to characterize the structural features of *Leishmania donovani* Xaa-Pro dipeptidase (LdXPD) and to identify potential small-molecule inhibitors. Recombinant LdXPD was successfully expressed and purified, and gel filtration chromatography confirmed its dimeric nature in solution. Spectroscopic analyses showed that the enzyme undergoes pH-dependent structural changes, with the highest conformational stability observed near physiological pH, while acidic and alkaline conditions induced noticeable rearrangements in both secondary and tertiary structure. Structural modeling highlighted the conserved overall fold of XPDs, including a well-defined metal-binding pocket and a stable dimeric interface similar to those seen in other species. The model provided the basis for virtual screening of the ConMedNP natural product library, leading to the identification of two promising candidates, Comp1032 and Comp11, with higher predicted binding affinity toward LdXPD than the human enzyme. Docking results indicated that both compounds form strong and persistent interactions within the catalytic cavity. Molecular dynamics simulations further supported these findings, showing stable enzyme-inhibitor complexes with minimal RMSD deviations and steady Rg values, confirming overall compactness. RMSF analysis indicated low fluctuations in catalytic residues, consistent with stable binding of inhibitors with LdXPD. SwissADME evaluation demonstrated that both molecules satisfy major drug-likeness criteria and possess favorable physicochemical and absorption properties. Overall, this work provides a detailed structural and dynamic assessment of LdXPD and highlights two natural-product-based inhibitors with promising drug-like potential, offering a foundation for developing selective anti leishmanial therapeutics.

Keywords: *Xaa-Pro dipeptidase, Virtual screening, Molecular Dynamics Simulation*

Computational insights into the antagonist-like effects of fatty acids on Estrogen Receptor- α : implications for drug discovery

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Breast cancer is still a leading cause of death worldwide, with the majority of instances being estrogen receptor-positive (ER $^{+}$). The estrogen receptor alpha (ER α) is a possible therapeutic target because it is known to play a crucial role in the development of tumors. In this study, we investigated whether a particular class of saturated and unsaturated fatty acids could act as ER α modulators. The current study used an in-silico method that integrated molecular docking, ADMET predictions, molecular dynamics (MD) simulations, and MM/PBSA free-energy calculations.

Based on docking and ADMET scores, six fatty acids were selected from a total of 72 with the potential to modulate ER α (PDB: 3ERT) for further molecular dynamics (MD) simulation studies. Stearic acid consistently emerged as the most promising option among the examined fatty acids. During MD simulation experiments, it demonstrated the lowest solvent-accessible surface area, compact receptor stability, minimized backbone fluctuations, and excellent docking affinity. Stearic acid was also evaluated as having the best relative binding free energy by MM/PBSA study. A promising set of interactions with ER α was demonstrated by other fatty acids, including nisinic acid and tetracosapentaenoic acid. However, they also promoted greater receptor flexibility and solvent exposure. Overall, these results suggest that fatty acids, particularly stearic acid, act as antagonist-like modulators of ER α by destabilizing helix 12 and hindering coactivator recruitment. These preliminary computational findings highlight stearic acid as a promising nutraceutical scaffold that may serve as a potential template for the development of new ER α -targeted antagonists.

Keywords: Breast cancer, Activation Factor 2, Molecular Simulation and Dynamics

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Candida albicans is an opportunistic fungal pathogen residing in warm-blooded animals that causes invasive and systemic infections under favourable conditions. It is one of the major leading cause of infectious diseases with only three major classes of antifungals in practice. The emerging virulence and rise of multidrug resistant *C.albicans* strains underscores immediate need for new, fungus-specific therapeutic targets. The fungal cell wall is made up of chitin, β -glucans and mannans. The several virulence mechanisms are associated with this dynamic cell wall composition and architecture. *PHR1* encodes a pH regulated β -1,3-glucanase (Phr1p) which plays a prominent role in maintaining cell wall integrity by remodelling of β -1,3-glucans, whose pH-dependent switching ensures survival of host in different niches. Gene disruption studies on *PHR1* of *C.albicans* reveals its role in adhesion and invasion by hyphal growth and morphogenesis. As a glycosylphosphatidylinositol-anchored extracellular enzyme lacking mammalian homologs, Phr1p represents an attractive target for selective antifungal development. However, no small-molecule inhibitors have been reported despite decades of target validation.

In this study, we employed structure-based virtual screening against Phr1p, built using Gas2 of *Saccharomyces cerevisiae* as template by homology modelling. A curated library of drug-like compounds from ChEMBL was prepared by applying stringent filters to enrich key pharmacophore features by excluding carbohydrate-based structures (have poor oral absorption) and including aromatic heterocycles capable of π - π stacking and ionizable groups facilitating cation- π interactions, based on insights derived from Gas2-ligand co-crystal structures. Finally, twenty structurally diverse compounds were selected for subsequent molecular docking and simulation studies. The top-ranked models followed favourable ADMET properties are predicted to form key interactions with catalytic residues Glu169 and Glu270. This work constitutes the first systemic effort to discover drug-like *PHR1* inhibitors, generating lead compounds for developing a novel antifungals targeting cell wall remodelling.

Keywords: *Candida albicans*, *PHR1*, structural-based virtual screening

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Motivation and Aim: Alzheimer's disease is a neurodegenerative disorder, characterized by progressive cognitive decline and loss of memory. Current therapies have limited efficacy, and less selectivity, so it is very important to look for new treatment strategies. CLK1, a kinase protein involved in RNA splicing and cell cycle regulation, is an attractive drug target for AD therapy. This study shows that cinchonine (CN) and resveratrol (RS), known to have neuroprotective properties, can inhibit CLK1 activity and, therefore, the progression of AD through a multidisciplinary approach.

Methods and Algorithms: Spectroscopic and calorimetric techniques were employed to measure the binding affinity, its mechanism and structural changes of CLK1 upon binding with CN and RS. Molecular docking studies were carried out to predict the binding modes and identify key interactions between the compounds and CLK1. Molecular dynamics simulations analyzed the molecular binding dynamics of the protein-ligand complex. Enzyme inhibition assays were carried out to determine the inhibitory potency of CN and RS against CLK1. The cytotoxic effects of CN and RS were evaluated in SH-SY5Y neuroblastoma cells using the MTT assay.

Results: Both of them, CN and RS showed a very high affinity with the *CLK1* as confirmed by spectroscopy and molecular docking studies. Molecular dynamics also showed the stable interactions of the free compounds, and the key residues present in the CLK1 pocket. Enzyme assays confirmed the reduced activity of CLK1 having IC₅₀ values of 8 μ M and 10 μ M to inhibition of CLK1 for CN and RS respectively. Further, The IC₅₀ values were found to be 205.6 μ M for CN and 73.5 μ M for RS on SHSY5Y cell line.

Conclusion: The findings of this investigation propose that targeting CLK1 with natural compounds like CN and RS can be a good way to deal with AD. Further research is still needed for clinical trials to validate these findings and explore the potential therapeutics of these compounds for the treatment of AD.

Keywords: Alzheimer's disease, cinchonine, resveratrol, spectroscopy, enzyme inhibition

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Targeting iron homeostasis Curtails mosquito's fertility and *Plasmodium*'s transmission potential

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Iron homeostasis is vital for mosquito fertility and for the growth of *Plasmodium* inside the vector, yet this biological connection has not been thoroughly investigated as a strategy for malaria intervention. Ferritin, the principal iron storing protein and a key regulator of cellular iron balance, remains largely uncharacterized in blood feeding insects. Our computational and evolutionary analyses show that mosquito ferritin is unusually configured, carrying a blend of conserved iron binding features together with unique haem coordinating residues that give it remarkably strong haem affinity, exceeding that of bacterioferritin. Silencing the ferritin heavy chain subunit in adult *Anopheles stephensi* females resulted in marked impairment of ovarian and associated tissue development, demonstrating its crucial involvement in mosquito physiology and in iron storage and distribution. Recent evidence that ferritin is also present in the *Plasmodium* food vacuole indicates a direct point of interaction between host and parasite centred on iron uptake. This opens the possibility of using ferritin mediated targeted delivery of antimalarial compounds into parasite compartments. By taking advantage of the competition for iron, we tested a panel of chelators and identified β -thujaplicin as a strong transmission blocking candidate that significantly reduced oocyst formation. Collectively, our results establish mosquito ferritin as a distinct and essential protein and emphasize iron regulation as a promising dual target pathway for future transmission blocking strategies, including ferritin based targeted drug delivery.

CONFERENCE ON EMERGING TRENDS
IN STRUCTURAL BIOPHYSICS

Investigating the dynamic role of MCM7 in Alzheimer's disease and breast cancer using bioinformatics approaches

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Cancer and neurodegenerative disorders are one of the most leading causes of deaths worldwide. The purpose of this work was to develop novel therapeutic strategies against these disorders. Breast Cancer and Alzheimer's Disease were selected as representative disorders to validate this relationship by conducting different bioinformatics approaches. The microarray datasets for the respective disorders were selected from Gene Expression Omnibus (GEO) database. The dataset code for breast cancer was GSE42568 and for Alzheimer's Disease, it was GSE33000. The common genes across the datasets were sorted using Venny webserver. It was found that CD209 (also known as Dendritic Cell-Specific Intercellular adhesion molecule-3-Grabbing Non-integrin or DC-SIGN) and MCM7 (Minichromosome Maintenance Complex Component 7) were the common genes. When the fold-change gene expression was investigated from the dataset, it was found that CD209 was upregulated in both the disorders, whereas MCM7 was downregulated in breast cancer but upregulated in Alzheimer's Disease. MCM7 is a family of replicative helicases that are involved in DNA replication. Upon analysing the structure of MCM7, it was found that these proteins are evolutionarily related to the AAA+ superfamily. They contain the MCM family domain, which includes motifs and that are required for ATP hydrolysis (such as the Walker A and B, and R-finger motifs). This makes for a promising target for therapeutic interventions against Breast Cancer and Alzheimer's Disease. Hence, it was established through this study that there is an inverse relationship between both these disorders.

Keywords: Cancer, Neurodegeneration, MCM7

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Carbon monoxide dehydrogenase (CODH)-expressing microbes convert the majority of the in situ generated carbon monoxide (CO) to carbon dioxide (CO₂), significantly reducing the ocean-to-air CO flux. Although a few CODH-expressing marine microbes have been characterized, the overall diversity and distribution of these CO-utilizing organisms across the oceans remain largely unexplored. In this study, we analyzed protein databases to identify marine genomes containing CODH-encoding genes. CODHs are of two types based on the active site metal content: the nickel-containing, oxygen-sensitive Ni-CODH, and the molybdenum-copper-containing, oxygen-tolerant MoCu-CODH. Our analysis identified Ni-CODH and MoCu-CODH genes from marine prokaryotes across various oceanic zones. Most habitats showed preferences for specific CODH types; for instance, the hydrothermal vent carboxydrotrophs predominantly have Ni-CODH genes, while most host-associated CO-metabolizers contain MoCu-CODH genes. Through phylogenetic analysis, Ni- and MoCu-CODHs are classified into seven and four distinct clades, respectively, revealing their evolutionary diversity. Genome context analysis showed that approximately 50% of Ni-CODH-encoding genomes contained genes coding Wood-Ljungdahl pathway proteins, while most ($\approx 70\%$) MoCu-CODH genes are in the neighbourhood of quinone-binding membrane protein. This study provides an overview of CO metabolism in marine microbes and its importance in oceanic ecosystems.

Keywords: *Metalloenzymes, Database analysis, Carbon Monoxide dehydrogenase*

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A molecule's ability to bind a receptor depends on the three-dimensional arrangement of key interaction features such as charged groups, hydrogen-bond donors/acceptors, and aromatic or hydrophobic elements, collectively known as a pharmacophore. Pharmacophore-based drug design aims to identify and prioritise these features to screen or design compounds with improved binding characteristics. Our objective is to develop a web tool capable of identifying molecules with a stronger binding affinity to a receptor than an existing reference drug. Using a known protein–ligand complex, one can evaluate and prioritize the interaction features that can contribute most strongly to binding, thereby helping to identify better drug candidates. Based on this concept, we developed PharmHIT.

PharmHIT operates using two pipelines. The first generates pharmacophore models by identifying ligand and protein features, mapping interactions within 5 Å, computing interaction scores, and normalising the summed contributions per ligand feature. It also modifies the ligand features to complement with the receptor that could enhance the binding by strengthening key pharmacophoric interactions. Based on user-defined model size (3–5 features) and number of models (1–9), the pipeline produces prioritised pharmacophore templates that retain high-scoring interaction features while resolving conflicts through a priority rule. These models can be visualised and exported for virtual screening. The second pipeline docks the candidate compounds retrieved from ZincPharmer against the target receptor and reports the binding affinities.

PharmHIT can identify the features that are most crucial for receptor recognition and optimizes for higher interaction energy. It can also guide researchers toward new compounds that mimic the strongest interactions of a known binder, allowing them to search for promising hits more effectively.

Keywords: *Pharmacophore models; pharmacophore features; protein-drug interactions.*



Poster Presentations

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IN STRUCTURAL BIOPHYSICS

Phosphorylation associated binding effects of p16 towards CDK4

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p16 is CDK (cyclin-dependent kinase) inhibitor, expressed by the p16INK4A belongs to the INK4 protein family. It functions as the tumor suppressor and highly vulnerable to cancer associated mutations and pathogenesis. Additionally, serves as biomarker in HPV-associated tumors, senescence, and aging when it is overexpressed. p16 specifically binds to the Cyclin D binding CDKs (active CDK), reducing the accessibility for cyclin binding consequently disrupting the interaction. This blocks the cell cycle progression from G1 to S phase, making p16 a crucial for cell cycle regulation. It's loss of function leads to uncontrolled cell proliferation and tumorigenesis. p16 is a compact structure and consists of 156 amino acids. It has four ankyrin repeats arranged in helix-turn-helix fashion. Ankyrin is stabilized by intra and inter hydrogen bonds and salt bridges at the interface of the p16 and CDK4, few of them were reported. PTMs play crucial role in proper functioning of a protein. It results in modulation of gene regulation, cellular signaling and (PPI) protein-protein interactions. Its effect changes pertaining to the residue type and position. Phosphorylation is one of the most common PTMs and has distinct effect on p16 functioning. By far, the PTMs and their impact on p16-CDK4 binding which are reported are minimal. Yet there are many phosphorylable residues are present in the p16 such as serine, threonine, and tyrosine. The study focuses on the effect of phosphorylable residues in p16 and CDK4 binding, which is further narrowed down to the phosphorylation occurs at N-terminal, intermediate and C-terminal residues. We are performing several analysis methods to examine the binding energy, complex stability, interresidue interaction, conformational analysis of the PTM added p16 in the complex.

Keywords: Pathogenesis, ankyrin, tumorigenesis.

Structural insights into cytokine cross-talk: computational analysis of Gpil-21 interactions with Gpil-4 And Gp IFN-Gamma

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Guinea pig (gp) is one of the most relevant small animal models for studying human tuberculosis and several other infectious and non-infectious diseases. Given this relevance as the interplay of guinea pig cytokines is crucial in combating TB infection, we aim to decipher their role by selecting the most important guinea pig cytokines for structural insights into cytokine cross-talk. Although this is an essential phenomenon for the regulation of immune responses, for most cytokine families, a structural basis for these interactions is still lacking. Interleukin-21 (IL-21) is a multifunctional cytokine operating at the junction between pro- and anti-inflammatory pathways, influencing both innate and adaptive immunity. A 3D structure model of gpIL-21 was thus built using homology modeling and was used as an anchor for protein-protein docking in order to predict possible binding surfaces with gpIL-4 and gpIFN-gamma. The results of docking allow us to see the structural relationships between gpIL-21, gpIL-4, and gpIFN-gamma and identify the putative binding sites, which provide detailed insight into the molecular aspects of cytokine cross-regulation—a step toward understanding the targeted control of the immune system during inflammation and infection. Further molecular dynamics simulations will be carried out for more detailed stability and structural analysis of these complexes. This computational study highlights the significant relevance of structure-based analysis toward deciphering gp cytokine cross-talk and should form the basis for future experimental validation of these models.

Keywords: *Guinea pig, Interleukin-21, Molecular Dynamic Simulations*

Systems biology based mechanistic modeling of hypoxia and cell cycle interactive dynamics in cancer cells

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Cancer progression is influenced by dysregulated hypoxia response and cell cycle control. Hypoxia stabilizes HIF1A, promoting survival and growth arrest, while MYC promotes proliferation. Their interplay, particularly through the regulation of p21 (CDKN1A), forms a critical feedback network influencing cancer cell fate. This study presents a mechanistic mathematical model integrating hypoxia signalling with cell cycle regulation using ordinary differential equations (ODEs). The model captures the dynamic interactions among HIF1A, MYC, p21, Cyclin D, RB, E2F, and Cyclin E at the cross regulation of cell cycle and hypoxia in cancer cells. Mathematical modeling, Simulations and Analysis were performed using XPPAUT. Results demonstrated bistability in the HIF1A–MYC network motif, enabling cells to switch between proliferative and arrested states. Under hypoxia, the system consistently reached a strongly stable G1-arrested state characterized by high HIF1A and p21, and suppressed MYC and E2F levels. Subsequently, cBioPortal based analysis across five cancer types performed revealed that the genetic alterations in these regulators are common, with hepatocellular carcinoma showing a significant link to poor prognosis. Further, Enrichr-based gene ontology analysis confirmed the network's role in G1/S transition, CDK regulation and transcriptional control. The integrated model developed highlights the regulatory significance of the HIF1A–MYC–p21 axis and its potential as a therapeutic target in hypoxia-adapted cancers.

Keywords: *Hypoxia, Cell Cycle, ODE Model, Cancer*

Machine learning-guided identification of structural vulnerabilities in Drp1: linking AI-predicted dynamics and evolutionary conservation to AML chemoresistance

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Acute Myeloid Leukemia (AML) remains one of the most aggressive hematological malignancies, with survival outcomes severely limited by therapy relapse and mitochondrial-mediated chemoresistance. Mitochondria undergo constant fission and fusion processes to maintain cellular fitness, but in AML this balance is disrupted. Hyperactivation of Dynamin-related Protein 1 (Drp1), a key mechano-chemical enzyme responsible for mitochondrial fission, drives excessive mitochondrial fragmentation. This morphologic shift prevents apoptosis, enhances oxidative stress adaptation, and enables leukemic cells to evade frontline chemotherapeutic stress. Therefore, targeting Drp1 regulation presents a promising strategy to restore apoptotic function and improve treatment sensitivity in AML.

We developed an integrated computational pipeline to investigate the molecular features driving Drp1 hyperactivity and reveal structural vulnerabilities for therapeutic targeting. A high-confidence AI-predicted 3D structure of human Drp1 was analyzed to assess domain organization and functional conformations. Normal Mode Analysis (NMA) was performed to identify low-energy motions linked to oligomerization, GTPase activation, and membrane constriction. Additionally, evolutionary conservation profiling across the Dynamin family highlighted functionally essential, highly constrained regulatory residues.

To translate these multi-parameter features into predictive capability, we developed a Machine Learning classification model trained using integrated structural flexibility, essential motion features, and conservation indices. The model is designed to predict unannotated regulatory sites, including conformational hinge regions and potential allosteric pockets capable of modulating protein activation. The expected outcome of this research is the generation of a robust structural-evolutionary signature of Drp1 regulation along with a validated ML-driven predictive map of novel druggable intervention points. Collectively, this computational discovery framework will assist in identifying lead inhibitory strategies that could reverse mitochondrial fragmentation, trigger apoptosis, and ultimately resensitize AML blasts to standard chemotherapy. This work supports the development of next-generation precision therapeutics targeting mitochondrial dynamics in cancer.

Keywords: Drp1, AML, Machine Learning

Structural characterization of *Candida Albicans* IDH1-IDH2 complex through integrative modelling and molecular dynamics simulations

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Candida albicans is an opportunistic fungal pathogen that lives as a commensal in healthy individuals and causes a wide spectrum of infections in immunocompromised individuals. Its high prevalence and significant mortality rates emphasize the need for better understanding and management of *Candida* infections. Isocitrate dehydrogenase (IDH) is one of the key enzymes of Tricarboxylic Acid (TCA) cycle, a fundamental metabolic pathway crucial for generating energy and synthesizing other biosynthetic precursors, that are essential for the survival and pathogenicity of an organism. It catalyzes the conversion of isocitrate to α ketoglutarate, linking carbon metabolism to energy production. IDH is an octameric complex formed by four heterodimeric complexes of IDH1 and IDH2, whose precise structure and regulatory mechanisms are not fully studied. In this study, the subunits IDH1 and IDH2 are modelled individually using Robetta and assembled into a heterodimer using HDock. The docked complex is subjected to molecular dynamics simulations in GROMACS to refine the interface and assess structural stability under near-physiological conditions. This equilibrated dimer is used for molecular docking of isocitrate (substrate for IDH) and selected effector molecules to rule out the potential substrate binding pockets and effector induced conformational changes. After docking, the stability and structural rearrangements in the enzyme-substrate complex are investigated using MD simulations. The inter-subunit interactions between IDH1 and IDH2, as well as the enzyme substrate interactions were analyzed using interaction analysis tools like LigPlus, to identify the key residues likely to control binding affinity and specificity. This study provides a detailed structural and functional framework for understanding substrate recognition and allosteric regulation of IDH in *C.albicans*. The identification of key residues involved in subunit interactions and enzyme-substrate interactions offers valuable insights for mutagenesis studies and developing targeted antifungal strategies.

Keywords: *Candida albicans*, Tricarboxylic cycle, Isocitrate dehydrogenase

Design and evaluation of novel peptidomimetic inhibitors targeting BCR-ABL tyrosine kinase in chronic myeloid leukemia

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Chronic Myeloid Leukemia is a type of blood cancer characterized by increased numbers of leucocytes. It originates from a chromosomal break and joining error, known as Philadelphia chromosome, in hematopoietic stem cells. A reciprocal translocation between chromosome 9 and 22 results in the fusion of BCR and ABL, encoding a constitutively active tyrosine kinase. As a result, the HSC is transformed into a leukemic stem cell that populates both myeloid progenitors and mature granulocytes (neutrophils). The disease progresses from an initial chronic to an accelerated phase and ultimately to a blast phase. Current available therapies include disease monitoring and TKI administration which have reduced progression to the advanced phase. However, reports of TKI resistance with drugs such as imatinib, emphasizes the need for alternative therapeutic strategies. The following study designs peptidomimetic inhibitors that bind to the substrate binding site of BCR-ABL Tyrosine kinase, contrary to the ATP binding site. Peptidomimetics were designed using sequences from MCoTI-13, which belongs to the MCoTI series created by grafting Abltide, the optimal ABL kinase substrate, onto the cell-penetrating cyclotide MCoTI-II. The work proposes that modification of loop 6 abltide sequence of MCoTI-13 can inhibit T315I mutation in ABL which is responsible for multidrug resistance. In addition to the cell-penetrating ability and stability conferred by the MCoTI-II scaffold, it is proposed that modification of loop 6 will enhance target specificity and inhibition of BCR-ABL tyrosine kinase, thereby reducing CML progression.

Keywords: *Peptidomimetic inhibitors, BCR-ABL Tyrosine Kinase, TKI Resistance*

Developing a comprehensive molecular interaction map for colorectal cancer metastasis

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Colorectal cancer (CRC) ranks as the second most prevalent malignancy globally, and it is the fourth leading cause of cancer-related mortality. Epithelial-to-mesenchymal transition (EMT) is an important developmental mechanism which is hijacked during CRC progression and metastasis. Metastasis is a complex multistep process which involves multiple signaling pathways. In this study, we present a comprehensive molecular interaction map of metastatic progression in CRC. The map is constructed based on an extensive literature survey of cell line-based data of CRC. The disease map integrates 120 different molecular regulators interconnected by 168 reactions, encompassing critical pathways such as TGF- β , PI3K/Akt, Wnt, Notch and JAK/STAT. The map was developed using CellDesigner adhering to SBGN and SBML standards. Subsequently converted into a logical modeling framework via CaSQ, and subjected to dynamic analysis using Cell Collective and GINsim platforms. These analyses elucidate the complex regulatory dynamics of EMT. Overall, the map developed and the subsequent analyses offer deeper understanding of the complex regulatory network governing the process of EMT in CRC.

Keywords: *Colorectal cancer, EMT, metastasis*

Analysis and insilico discovery of driver genes and pathways facilitated by Mapk4 in cancer progression: a graphical approach

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MAP Kinases fall under the family of Serine Threonine Kinases- very well known for their role in regulating cellular functions- activated through external factors and signals like stress, growth, etc relay signals from the external cellular surface to the nucleus aiding in regulating processes like cellular proliferation, differentiation, survival mechanisms and cell death/ induced apoptosis. As an atypical member of the MAPK family, MAPK4 has been relatively understudied with respect to it's pathways (due to its atypical activation loop), signalling cascades and very recently, it's role in cancer progression. In this experimental study, the TCGA pan-cancer

expression ($\approx 20k$ genes, $\approx 11k$ samples) was integrated with a MAPK4-centred STRING PPI

graph (646 nodes, 31,575 edges) to help identify six communities, two of which were strongly enriched and expressed in MAPK-related pathways and were thus, used as labeled signalingmodules. A two-layer GCN trained on expression and network features of the two communities identified achieved a test ROC AUC of 0.656 and AUPRC of 0.340, and prioritized 226 genes as candidate MAPK4-associated signaling nodes. The highest-confidence predictions included high-centrality hubs such as ADCYAP1, ADCY9, ACVR1B, ACP1, and ADAM10, suggesting potential novel intermediates and crosstalk partners

Keywords: *Graph convolution networks (GCN), cancer signalling genes and pathways, MAPK4*

Structural determinants for phenotypic diversity in SCN5A-associated cardiac diseases

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The assembly of proteins into complexes and their interactions with other biomolecules are often vital for their biological function. Mutations at protein interfaces are known to have a high potential to be damaging and cause human genetic diseases, but there has been relatively little consideration for how this varies between different types of interfaces. Residues that do not directly participate in an interface, but are close in three-dimensional space, also show a significant disease enrichment.

The SCN5A gene encodes the alpha-subunit of the Nav1.5 ion [channel protein](#), a major voltage-gated sodium channel in the heart, which is responsible for the sodium inward current. Variants in SCN5A cause distinct genetic arrhythmia syndromes like Brugada syndrome and congenital long QT syndrome, which can cause sudden cardiac death. SCN5A variants have also been associated with isolated conduction system disease, atrial fibrillation, atrial standstill, sick sinus syndrome, sudden infant death syndrome (SIDS) and dilated cardiomyopathy.

Nav1.5 functions within a regulatory connexome comprising interactions with β -subunits, connexin-43, plakophilin-2, ankyrin-G, and a few others. Mutations disrupting these protein-protein interaction interfaces produce clinical phenotypes through independent mechanisms, yet their spatial localization at the connexome binding sites has not been systematically mapped to disease outcomes.

Using the human Nav1.5 cryo-EM structure, 100+ pathogenic SCN5A single nucleotide variants and missense mutations from ClinVar database, were spatially mapped onto the 3D model. Statistical enrichment analysis assessed whether interface localization correlates with cardiac phenotypes.

Significance: This work links mutations at connexome interfaces to functional consequence, providing a framework for phenotype prediction in unreported variants and identifying interface-specific therapeutic targets. This represents systematic integration of cryo-EM structural data with genotype-phenotype correlation, advancing mechanistic understanding of Nav1.5 channelopathies.

Keywords: SCN5A mutations, protein-protein interfaces, connexome disruption.

Immersive structural biophysics: AR/VR-based interactive visualization of protein structure, dynamics, and function

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Understanding how a protein's shape changes over time is key to explaining how it functions in a living system. However, most visualization tools still show these molecules as static objects. This makes it hard to understand their true behavior. In this work, we explore an interactive method that brings structural biophysics into an engaging environment using augmented and virtual reality. By placing protein models in 3D space, users can walk around a structure, see subtle shape changes, and follow dynamic processes like ligand binding or channel opening as if they were happening in front of them. This AR/VR-based framework aims to make molecular interactions easier to grasp, especially for complex systems where movement is crucial for function. It also opens up new possibilities for teaching, collaborative research, and quickly generating hypotheses since multiple users can examine the same molecular event from various angles. Our study shows how immersive visualization not only improves understanding of structure function relationships but also allows for more accessible and engaging biophysical analysis. This work proves that extended-reality tools can connect molecular data and human perception, promoting a more interactive and exploratory approach to studying biomolecules.

Keywords: *Augmented Reality (AR), Virtual Reality (VR), Immersive Visualization, Protein Structure and Dynamics, Structural Biophysics, 3D Protein Modeling, Molecular Interaction Analysis, Collaborative research, Structure Function Relationships*

CONFERENCE ON EMERGING TRENDS
IN STRUCTURAL BIOPHYSICS

Predictive modelling of autism spectrum disorder using WEKA: a comprehensive classifier performance study

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Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition requiring accurate and early identification to support timely intervention. In this study, we evaluated the predictive performance of multiple machine-learning classifiers implemented in the WEKA environment using a publicly available ASD dataset. Fourteen algorithms were applied to assess classification accuracy, precision, recall, F1-score, specificity, and error metrics, enabling a comprehensive comparison of model robustness. Exceptional performance was observed for Bagging, Logit Boost, Decision Table, PART, J48, and Random Forest classifiers, each achieving 100% accuracy, perfect precision, recall, F1-score, and specificity, indicating their strong ability to distinguish ASD from non-ASD cases without misclassification. Multilayer Perceptron also demonstrated outstanding predictive capability with 99.65% accuracy, high precision (0.9934), recall (1.000), and F1-score (0.9967). Naïve Bayes and SMO maintained strong performance with accuracies of 98.97% and 96.57%, respectively, while Logistic Regression produced moderate accuracy (95.20%). In contrast, Voted Perceptron and Random Tree showed reduced effectiveness, and the lowest predictive ability was documented for Stacking and Vote algorithms (51.71% accuracy), suggesting their inadequacy for ASD classification. Confusion matrix analysis confirmed the reliability of high-performing classifiers, with several achieving error-free prediction across ASD and non-ASD categories. Overall, the study highlights the suitability of ensemble-based and tree-based models for ASD prediction and underscores the potential of machine-learning approaches in strengthening automated screening frameworks for neurodevelopmental disorders.

Keywords: *Autism Spectrum Disorder, Machine Learning, WEKA, Classification Accuracy, Ensemble Methods, Predictive Modelling*

Predicting ulcerative colitis using 16S Rrna microbiome data through combined RNA structural analysis and machine learning

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Ulcerative colitis (UC) is a chronic inflammatory bowel disease characterized by persistent colonic inflammation, with gut microbiome alterations playing a critical role in disease onset and progression. The primary objective of this study is to classify UC and healthy individuals using 16S rRNA sequencing data through machine learning while integrating RNA secondary structure prediction to explore structural variations in microbial RNA associated with UC. Raw 16S rRNA sequences from UC patients and healthy controls were retrieved from the NCBI Sequence Read Archive and pre-processed to remove low-quality reads, ambiguous bases, and duplicate sequences. Feature extraction was performed using k-mer profiling, GC content, and nucleotide composition to quantify microbial patterns. To incorporate a structural biology component, RNA secondary structures were predicted using ViennaRNA RNAfold, enabling visualization of folding patterns and potential structural instabilities in UC-associated microbial RNA. Machine learning models including Random Forest, Naïve Bayes, and Logistic Regression were trained to classify UC versus healthy samples, with Random Forest achieving the highest accuracy. This integrated sequence- and structure-based approach highlights the potential of combining microbiome analysis with computational modeling to enhance UC prediction and provide functional insights into microbial dysbiosis associated with the disease.

Keywords: *Ulcerative Colitis, Gut Microbiome, RNAfold, Machine Learning, Naïve Bayes, Logistic Regression*

Benchmarking AI-predicted structures against experimental gold standards: a multi-pathogen comparative study on *M. Tuberculosis*, *P. Falciparum*, and SARS-Cov-2 variants

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The integration of Artificial Intelligence (AI) into structural biophysics has revolutionized the modelling of drug targets. However, the reliability of tools like AlphaFold2 varies across different pathogen classes and requires rigorous validation against experimental data, particularly for emerging variants. This study evaluates the structural accuracy of AlphaFold2 by benchmarking its predictions against high-resolution X-ray crystallographic data for three distinct pathogen categories: bacterial (*Mycobacterium tuberculosis*), protozoan (*Plasmodium falciparum*), and viral (SARS-CoV-2 Delta and Omicron variants). The amino acid sequences for *M. tuberculosis* RNA Polymerase β -subunit (rpoB), *P. falciparum* DHODH, and the Receptor Binding Domains (RBD) of SARS-CoV-2 variants were retrieved from UniProt and PDB. Three-dimensional models were generated using the AlphaFold2-mmSeqs2 pipeline via Google Colab. Structural fidelity was assessed by superimposing these models onto their respective experimental structures (PDB IDs: 5UHC, 6GJG, 7V7Q, 7QO7) using the TM-Align algorithm. The analysis revealed a clear dichotomy in AI prediction accuracy. The rigid metabolic enzyme Pf-DHODH showed near-atomic convergence with the crystal structure (RMSD: 0.56 Å; TM-Score: 0.99). In contrast, the *M. tuberculosis* rpoB complex exhibited a higher deviation (RMSD: 3.28 Å), attributed to the intrinsic flexibility of the clamp domain rather than misfolding (TM-Score: 0.91). Notably, the SARS-CoV-2 analysis showed that AlphaFold2 captured the global fold of both variants but struggled to resolve the hyper-mutated loops of Omicron (RMSD: 3.24 Å) compared to Delta (RMSD: 2.56 Å). This study demonstrates that while AlphaFold2 is a robust tool for stable enzymatic targets, it requires careful interpretation when applied to flexible molecular machines and heavily mutated viral proteins. These findings establish a cost-effective workflow for in silico structural validation in resource-limited settings.

Keywords: AlphaFold2, structural accuracy, SARS-CoV-2, *Mycobacterium tuberculosis*, *Plasmodium falciparum*

Integrated in-silico and preliminary in-vitro characterization of RVB1 and RVB2 in *Candida Albicans*

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RVB proteins, comprising the conserved paralogs RVB1 and RVB2, are AAA+ ATPases widely present in archaea and eukaryotes. Their ability to form the RVB1/2 complex and integrate into diverse macromolecular assemblies underlies key roles in transcription, chromatin remodeling, DNA repair, and stress responses. Although highly conserved, the structural and functional properties of the *Candida albicans* homologs remain poorly defined. In this study, we used an integrated in silico and in vitro approach to characterize CaRvb1 and CaRvb2. Primary sequence attributes were analysed using ProtParam, followed by secondary-structure prediction with SOPMA. Conserved domains were identified through the NCBI Conserved Domain Database (CDD), and multiple sequence alignment using ClustalW revealed strong conservation of classic AAA+ motifs, including the Walker A/B domains. Tertiary structures were predicted using both SWISS-MODEL and AlphaFold3, generating the structural models of CaRvb1 and CaRvb2. Exploratory in vitro experiments are currently being conducted to examine the phenotypic behaviour and stress-response profiles of RVB1 and RVB2 mutants in *C. albicans*, providing early functional insights that complement the computational predictions. Future work will involve modelling the CaRvb1-CaRvb2 complex, analysing ATP-binding pockets, predicting ATP docking and binding affinity. These studies will contribute to a deeper mechanistic understanding of RVB function in fungal biology. Overall, this combined in-silico and in-vitro investigation provides an initial structural and functional framework for RVB1 and RVB2 in *C. albicans*, laying the foundation for future genetic and mechanistic research on RVB proteins and their role in fungal pathogenicity.

Keywords: RVB1, RVB2, *Candida albicans*

Influence of bamboo leaf ash weight fraction on mechanical, microstructural, and tribological properties of Al2218-B₄C MMC

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The influence of agro-waste-derived reinforcements on the mechanical, microstructural, and tribological behaviour of aluminium metal matrix composites has become a significant research focus in the development of lightweight and sustainable engineering materials. Aluminium Metal Matrix Composites (AMMCs) are widely recognized for their excellent mechanical and tribological performance, making them suitable for automotive, aerospace, and marine components subjected to wear and stress. Incorporating suitable reinforcing particulates further enhances the capabilities of the base aluminium matrix. Al2218 alloy, selected for its excellent strength and wear resistance, serves as the matrix material. Bamboo Leaf Ash (BLA) improves mechanical behaviour, while Boron Carbide (B₄C) contributes to enhanced hardness and wear resistance. In this study, hybrid Al2218 composites were fabricated using the stir casting method with B₄C held constant at (2%, 4% and 6%) and BLA weight fractions of (1%, 3%, and 5%). The composites were evaluated for tensile strength, percentage of elongation, compressive strength, impact strength, hardness, and wear resistance. Wear tests were performed at sliding velocities of 1–3 m/s, a constant sliding distance of 2000 m, and an applied load of 10 N. Microstructural and phase analysis using Scanning Electron Microscopy (SEM) and X-Ray Diffraction (XRD) revealed uniform particle distribution, strong matrix–reinforcement bonding, and formation of interfacial phases. Porosity was assessed by comparing theoretical and experimental densities. The results show that increasing BLA content significantly enhances hardness, wear resistance, and overall mechanical performance of Al2218 composites. This highlights the effectiveness of agro-waste-derived BLA as a secondary reinforcement, making these hybrid composites highly suitable for advanced engineering applications.

Keywords: Al2218, B₄C, Bamboo Leaf Ash, Hybrid Metal Matrix Composites, Stir Casting.

In Silico* structural and functional characterization of cytolethal distending toxins from *Campylobacter Jejuni

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Campylobacter jejuni is a gram negative, motile, food-borne pathogen causes human gastroenteritis worldwide. Recent studies indicate that *C. jejuni*'s infection leads to campylobacteriosis with symptoms diarrhea, stomach pain, and fever which may result in serious post-infectious consequences such as Guillain-Barré and Miller-Fisher syndromes. Intriguingly, *C. jejuni*'s pathogenicity is mediated by a number of virulence factors, including flagellar motility, adhesion mechanisms, and the secretion of powerful toxins such as the Cytolethal Distending Toxins (CDT). These toxins disrupts cellular processes causing DNA damage, cell cycle arrest, and host apoptosis in host cells. Despite their central role in pathogenesis and inducing lethality, the structural and functional characterization of *C. jejuni* CDT's were unexplored. The current study sheds light on structural and functional properties of these toxins unraveling the conserved domains using computational tools. Moreover, the study provides insights into their molecular mechanism of toxicity by molecular docking and molecular dynamic simulations.

Keywords: *Campylobacter jejuni*, Cytolethal Distending Toxins, apoptosis, lethality, molecular docking, and simulations.

Identifying therapeutic targets in triple-negative breast cancer through AI profiling of YY1 signaling networks

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Raf Kinase Inhibitor Protein (RKIP) and Yin Yang 1 (YY1) are critical regulators of cancer progression, immunity, apoptosis, and inflammation. RKIP, a phosphatidylethanolamine binding protein family member, functions as a metastasis suppressor by inhibiting Raf/MEK/ERK, NF- κ B, and GPCR signaling pathways. It modulates MAPK, NF- κ B, STAT3, and PI3K/AKT cascades, suppresses invasion via MMP13 downregulation, limits tumor associated macrophage recruitment, and promotes miR-185 expression to inhibit HMGA2 and SNAI1, thereby blocking epithelial-mesenchymal transition. YY1, a multifaceted transcription factor, exhibits both oncogenic and tumor-suppressive functions across malignancies. In breast cancer, YY1 drives proliferation via LINC00673 mediated Notch activation and fosters an inflammatory tumor microenvironment through IL-6 elevation and Treg dysregulation. In triple-negative breast cancer (TNBC), we observed frequent epigenetic silencing of RKIP and overexpression of YY1, suggesting an oncogenic shift and an uncharacterized regulatory interplay. This study merges TCGA, ENCODE, and GEO datasets with machine learning algorithms, including XGBoost and Random Forest, to derive predictive signatures for metastasis and therapy response. RKIP-YY1 interaction networks, linked to EMT, tumor-associated macrophages, and Hippo pathway components, were constructed using STRING and Cytoscape. Target identification through DrugBank and ChEMBL, followed by virtual screening in Discovery Studio and identified Ebselen as a potential YY1 inhibitor. Together, these integrative computational and molecular analyses provide new insights into RKIP-YY1 driven vulnerabilities in TNBC.

Keywords: YY1, Machine Learning, Drug Discovery.

Integrative bioinformatics approach for epitope prioritization and vaccine construct design against *Pseudomonas Aeruginosa*

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Pseudomonas aeruginosa is an ESKAPE opportunistic pathogen that has the property of antibiotic resistance, causing nosocomial infections. Despite the increase in clinical diseases of *P. aeruginosa*, there is no effective vaccine available yet. Therefore, the current computational study aims to design a multi-epitope vaccine against *P. aeruginosa* using computational tools available for vaccine construction. A reference strain of *P. aeruginosa* was retrieved from the UniProt database. Cytotoxic T lymphocyte (CTL), helper T lymphocyte (HTL), and linear B lymphocyte (LBL) epitopes were identified using immunoinformatic approaches. These epitopes were then screened for antigenicity, allergenicity, and toxicity. Further, the vaccine construct was designed using the epitopes along with the adjuvant and appropriate linkers. The vaccine construct was analysed for the physicochemical properties, including stability, optimal molecular weight, followed by two-dimensional and three-dimensional structure prediction. Moreover, molecular docking of the vaccine construct was performed with Toll-like receptor 5 (TLR5) using ClusPro and has exhibited strong and stable interactions. Additionally, Immune simulation has indicated a long and strong immune response. Overall, this computationally designed vaccine candidate appears promising; however, further in vitro and in vivo validations are necessary to confirm its efficacy and immunogenic potential.

Keywords: *Pseudomonas aeruginosa*, Multi-epitope Vaccine, ESKAPE pathogens, Immuno-informatics, Drug resistance.

A living sensor that whispers about the seismic waves (Earthquake And Tsunami)

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Earthquake and Tsunami are the 2 most destructive natural calamities that we dealt from years. The most devastating problem is that (till now) they are not per-detectable. It is so because the stress accumulation deep inside earth rarely produces directly detectable signal before the rupture. This is a proposed research proposal to work on the production of the Microbial Pre-Seismic (MPS) sensor that investigates a noble hybrid sensing strategy that uses living microbial that shows abnormal behavior (such as concentration increase) to detect stable stress accumulation in the rocks inside the earth. Far inside earth's surface lies an immense and mysterious life support systems, which are very far from human to reach, that sustains without sunlight (which is the main spouse of It in the earth's crest), in high pressure, in extremum chemical experiments. Traditionally (till now) this deep life support system has been considered as isolated from the phenomena on the earth's crest. However, the tectonic movement establish the co-relation between these two different living kingdoms. What actually happens, during the stress accumulation deep inside earth, small micro-cracks are formed that releases dissolve gases such as hydrogen, methane and small particles of iron and sulphur, PH change spike of some redox gradients in the ground water in the sedimentary environment. These sudden shifts directly influence the metabolism of ecological environment of the micro-organisms. Specific microbial such as Thiobacillus, Desulfo-vibrio and many more respond rapidly to such shifts because their energy pathways depend on these chemicals. This anomalous change leads to changes in bimolecular concentration such as ATPs or extra cellular enzymes are expected to observe days to weeks before earthquake followed by Tsunami. This research proposed is to make a living MPS sensor using the microbial. All these are based upon one-line single sentences concept: gradual fatigue, chemical immigration and pressure accumulation through stress induced living microbial signatures. To advance this concept, we propose hybrid research frame work that combines real time hydro-chemical monitoring before the seismic waves. We termed this branch as seismo-bio-chemito-logy, that combines physics, biology and chemistry into a single unified topic. Both Thiobasillus, Desulfo-vibrio are bound commonly in water bodies and soil sediments, which are not that much difficult to culture and grow them. Continuous sampling of the water bodies and soil sediments can give us an expected biological activity spikes – days to weeks before the seismic waves. Some specific genes are also expected to be found in this environment some times before the seismic period. By treating these living kingdoms as a unified living brotherhood, we can get an dynamic, sensitive and densely distributed monitoring network spread within the lithosphere. Understanding this seismo-bio-chemito-logic interaction not only strengthens earthquake pre-deteating systems but who provide a profound insight, but also a new pathway how we relate to our brotherhood living Kingdom that whispers about the earthquake. All these are based upon one line single sentences concept: gradual fatigue, chemical immigration and pressure accumulation through stress induced living microbial signatures.

Keywords: Earthquake, Tsunami, Microbial, Microbes, sensor, Pre seismic sensor, seismic waves

PartnerIDP: Predicting Interaction-Driven Conformations of IDPs Using Machine Learning

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Intrinsically disordered proteins (IDPs) defy traditional structure prediction because of their extreme conformational flexibility and context dependence. Upon binding different partners, many IDPs experience disorder-to-order transitions instead of adopting a single stable fold, leading to partner-specific folded conformations. While AlphaFold3 and other general-purpose structure prediction models are excellent at modelling folded protein complexes, they are still not very good at capturing these partner-induced IDP conformations. These models are limited in their applicability to signalling, regulatory, and drug-discovery contexts where IDP function is determined by binding partners.

PartnerIDP is a deep learning framework created especially to predict bound conformations of IDPs conditioned on their structured interaction partners. From the literature and structural databases such as DIBS, IDEAL, Disprot and PDBsum, we assembled a high-confidence database containing IDP - interacting partner protein complexes that included IDP sequences, partner sequences, experimentally determined PDB structures, and interacting residues. ESM-2 embeddings are used to encode both IDPs and partners, and interface-aware cross-attention mechanisms or compact summary representations are used to incorporate partner context. The model predicts bound-state IDP coordinates.

Thus, PartnerIDP will exhibit improved interface geometry due to explicit partner conditioning and is believed to achieve an appropriate mean RMSDs, expected to remain comparable to AlphaFold3 baselines. Furthermore, partner-specific evaluations will directly capture disorder-to-order transitions driven by interaction context by revealing different predicted conformations for the same IDP when bound to different partners. Overall, this data-efficient computational model complements current structure prediction tools by addressing a key gap in IDP structural prediction and providing a scalable foundation for studying context-dependent folding mechanisms.

Natural anti -TB leads from *Neolamarckia Cadamba* reveals high affinity Inha binders with favourable ADMET signatures

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Tuberculosis (TB) caused by *Mycobacterium tuberculosis* (M.tb.), remains one of the leading causes of mortality worldwide, underscoring the urgent need for novel therapeutic strategies. This study aims to identify novel inhibitors of the InhA enzyme, which is firmly validated as a critical therapeutic target for tuberculosis, due to its essential role in mycolic acid biosynthesis- an indispensable component of the mycobacterial cell wall. We selected 34 active compounds from *Neolamarckia cadamba* belonging to *Rubiaceae* family for molecular docking analysis. The evaluation of druglikeness was performed based on Lipinski's Rule of Five, while the computed phytochemical properties and molecular descriptors were used to predict the ADMET of the compounds. Among these, 20 compounds were further examined by performing molecular docking analysis with InhA target using Autodock 4.2. The molecular docking analysis showed that naringenin, a major phytochemical from *Neolamarckia cadamba*, possess promising inhibitory potential against InhA, with binding energy of -7.78 kcal/mole. The high binding energy of naringenin compared to other phytocompounds and standard compound Isoniazid suggests its potential as a promising InhA inhibitor.

Keywords: *Tuberculosis, InhA, Naringenin, Molecular docking, Neolamarckia cadamba.*

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About the Conference

Structural biophysics is shifting from a static view of biomolecules toward an integrated understanding of their structure, dynamics, and function. Advances in dynamic structural biology—through techniques like time-resolved crystallography, NMR spectroscopy, cryo-EM, and molecular dynamics simulations—highlight the importance of molecular motions in biological mechanisms and drug discovery. At the same time, integrative and hybrid methods combining multiple experimental approaches are enabling the study of large, flexible, and transient biological assemblies with unprecedented detail. The field is also being transformed by machine learning (ML) and artificial intelligence (AI), with breakthroughs like AlphaFold redefining structure prediction and data analysis, though critical interpretation remains essential. Additionally, a new understanding of allosteric networks and energy landscapes reveals that protein function often arises from shifts across ensembles of conformations rather than discrete states. This expanded view is reshaping research priorities, training needs, and therapeutic strategies, urging academicians to integrate dynamic, multi-method, and computational perspectives into modern structural biophysics education and scholarship. This conference will serve as a vital platform to explore these emerging trends, equipping researchers, educators, and students with the knowledge and tools needed to navigate and lead in this evolving landscape. Through discussions, and expert insights, participants will gain a deeper appreciation of how dynamic, multi-method, and computational approaches are redefining the future of structural biophysics.

Department of Biotechnology

The Department of Biotechnology at NIT Warangal is a premier centre for innovation, research, and academic excellence, offering B.Tech, M.Tech, and Ph.D. programs in Biotechnology. Department of Biotechnology established in 2006 with the B.Tech in Biotechnology program under the TEQIP initiative, the department has continually expanded its academic offerings. The M.Tech program was introduced in 2021, and the Ph.D. program began in 2008. Our curriculum is meticulously crafted to meet the ever-evolving demands of the biotechnology industry and is regularly updated to align with the latest advancements. With 20 dedicated faculty members specializing in diverse domains of biotechnology, the department fosters a vibrant academic and research environment. Our state-of-the-art laboratories provide students with hands-on experience and access to cutting-edge technologies, ensuring their readiness for academic, research, and industrial challenges. The department has successfully executed numerous research projects and currently manages 10 ongoing research projects, funded by prominent agencies like DBT, DST, ICMR, and CSIR, under the Government of India. We actively support Research Associates through funding from various agencies and also undertake consultancy projects for industries, bridging the gap between academia and real-world applications.

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